

BFS Traversal

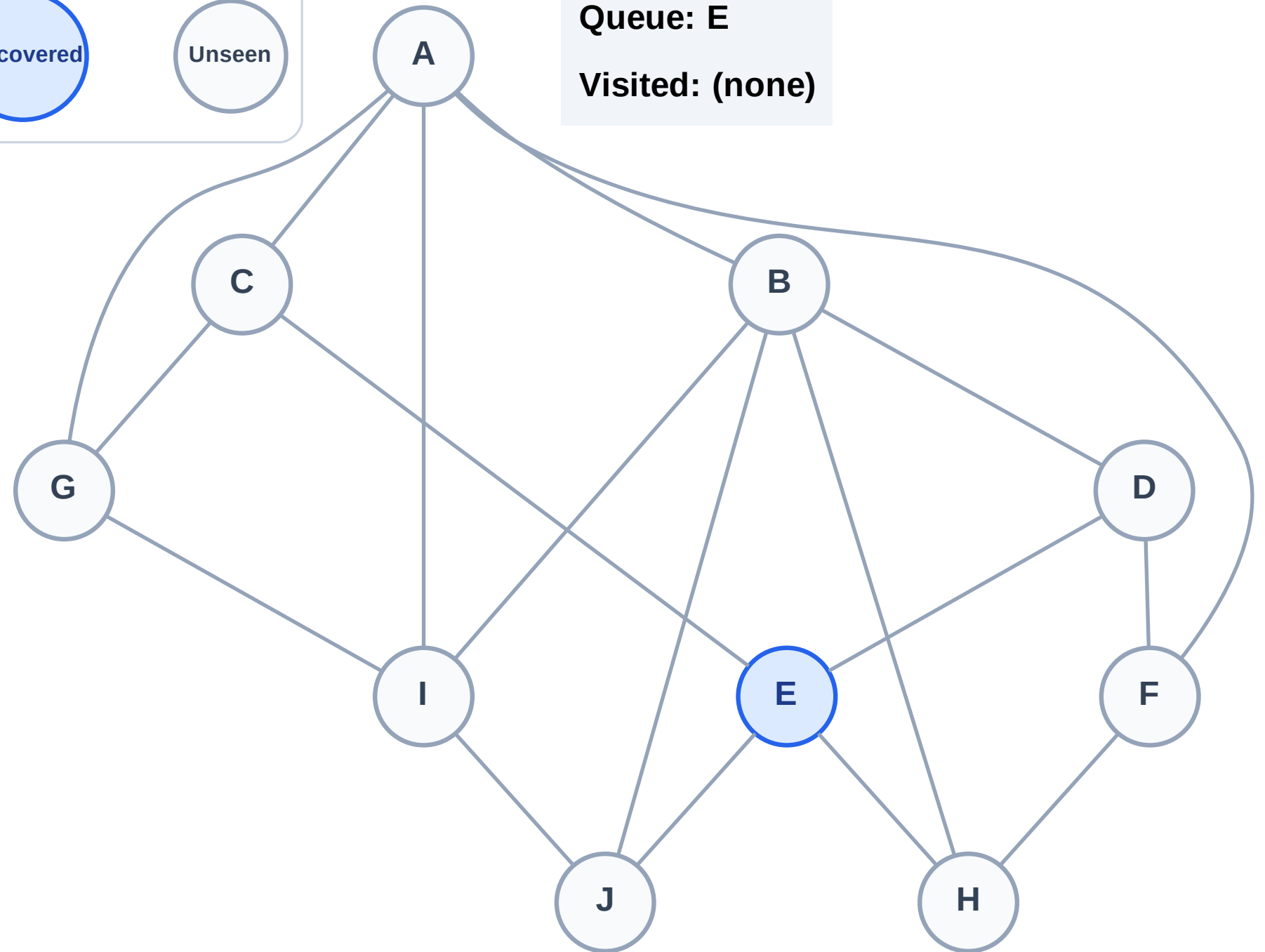
Step 1: Start at E

Legend



Queue: E

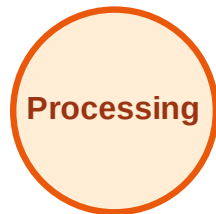
Visited: (none)



BFS Traversal

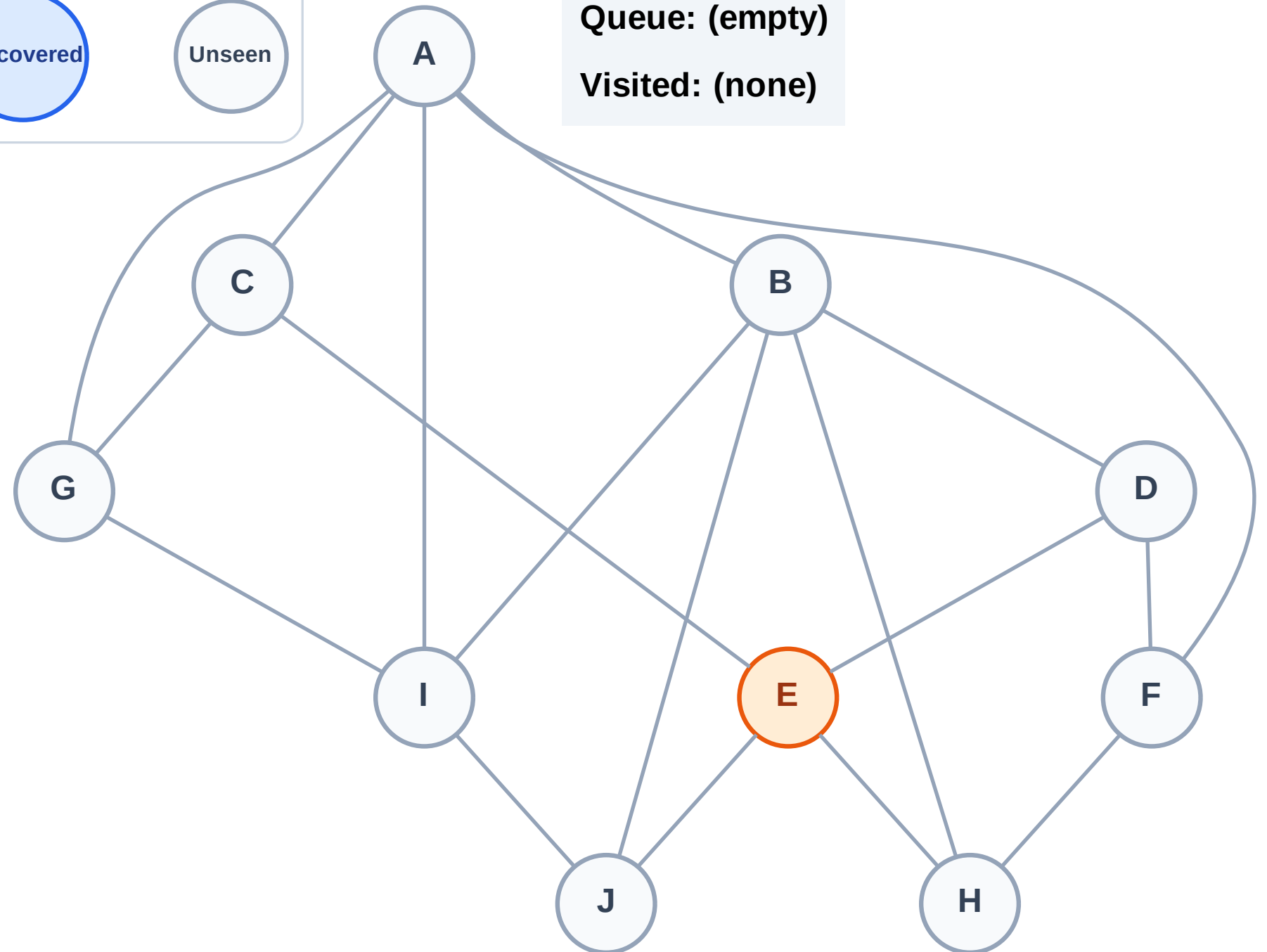
Step 2: Process node E

Legend



Queue: (empty)

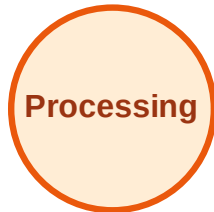
Visited: (none)



BFS Traversal

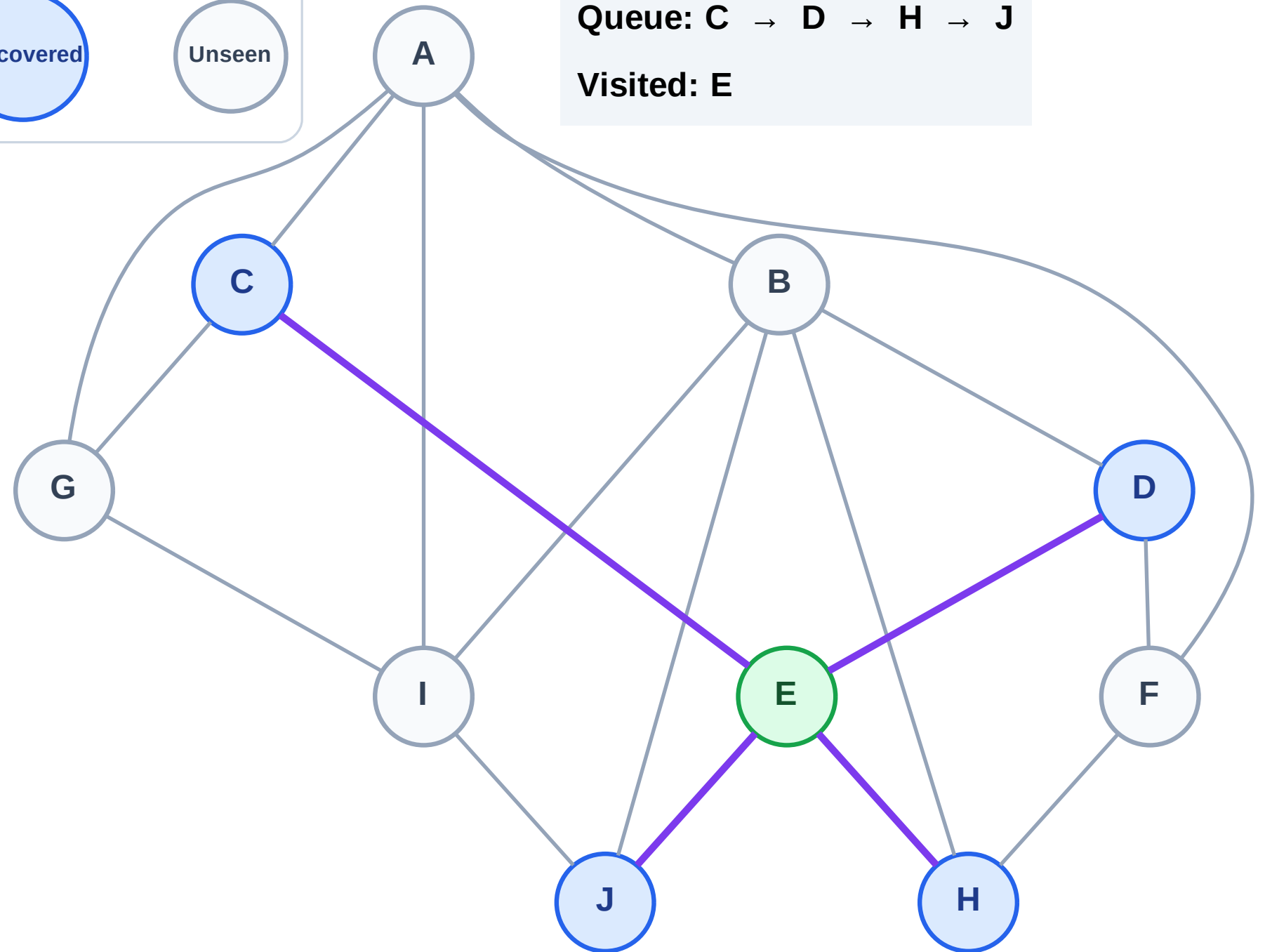
Step 3: Discover C, D, H, J from E

Legend



Queue: C → D → H → J

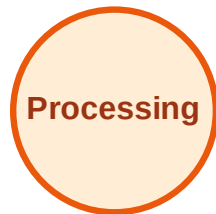
Visited: E



BFS Traversal

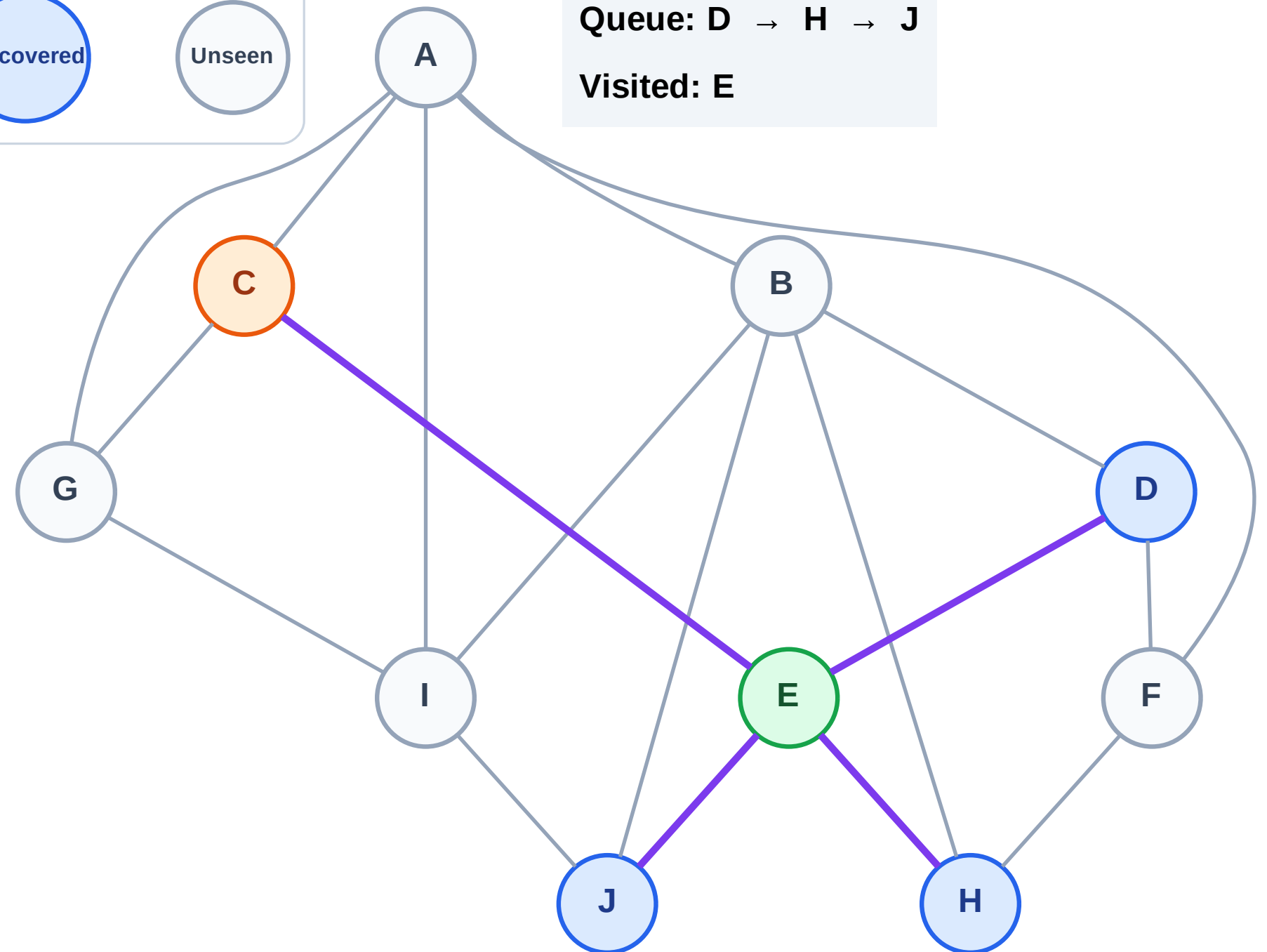
Step 4: Process node C

Legend



Queue: D → H → J

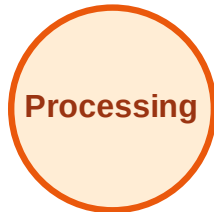
Visited: E



BFS Traversal

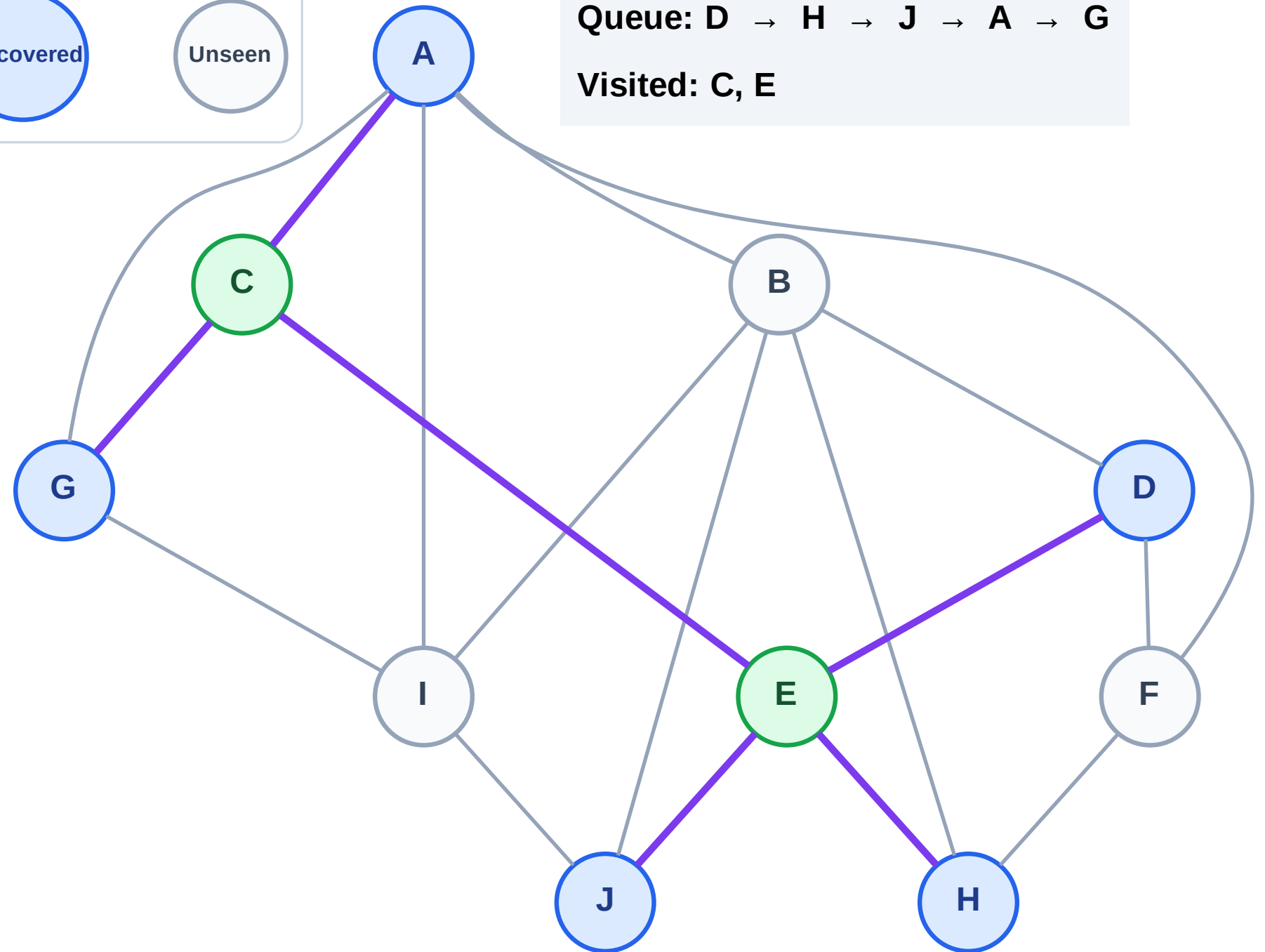
Step 5: Discover A, G from C

Legend



Queue: D → H → J → A → G

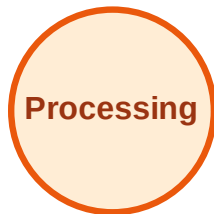
Visited: C, E



BFS Traversal

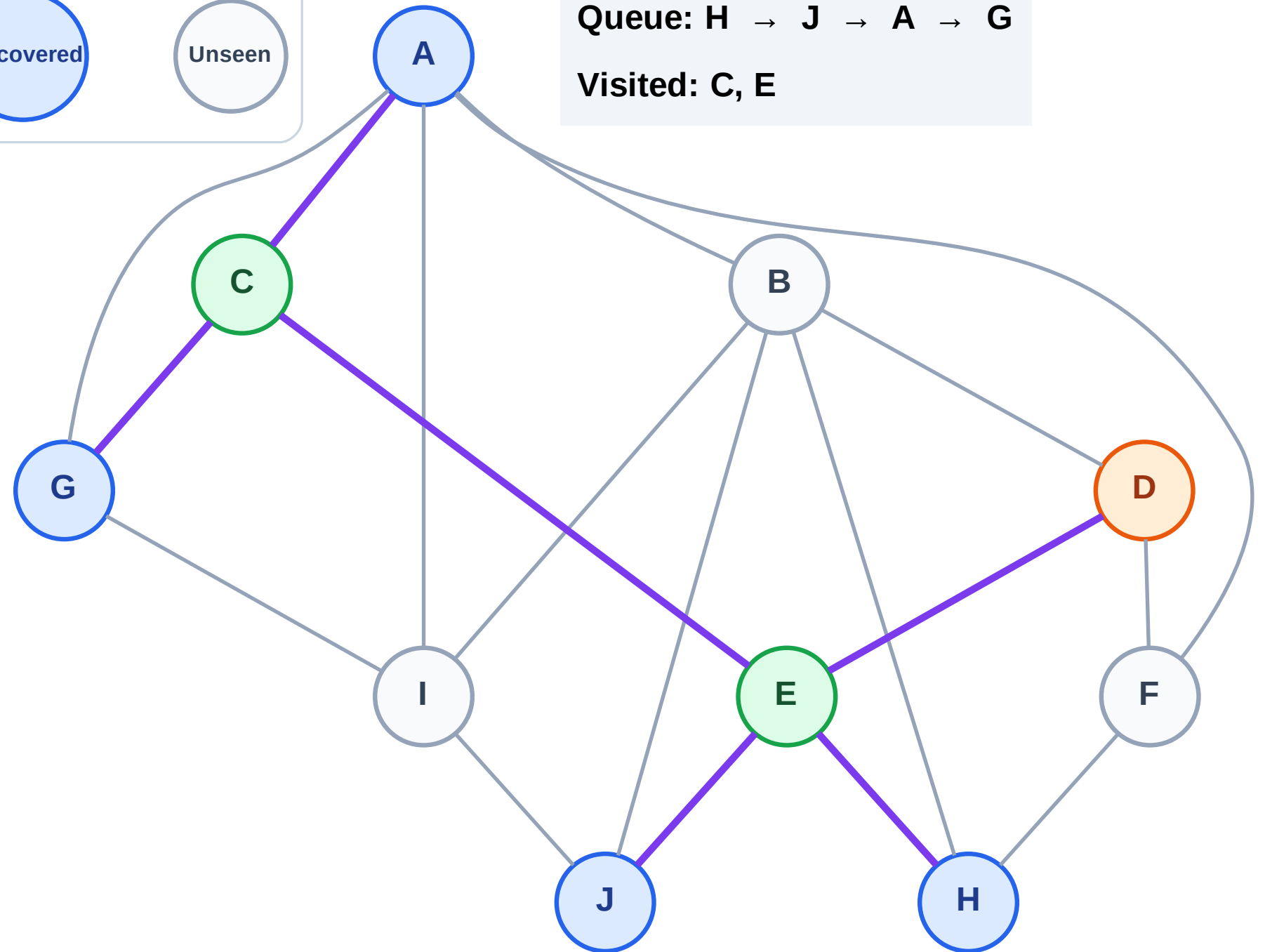
Step 6: Process node D

Legend



Queue: H → J → A → G

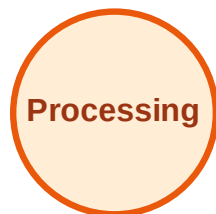
Visited: C, E



BFS Traversal

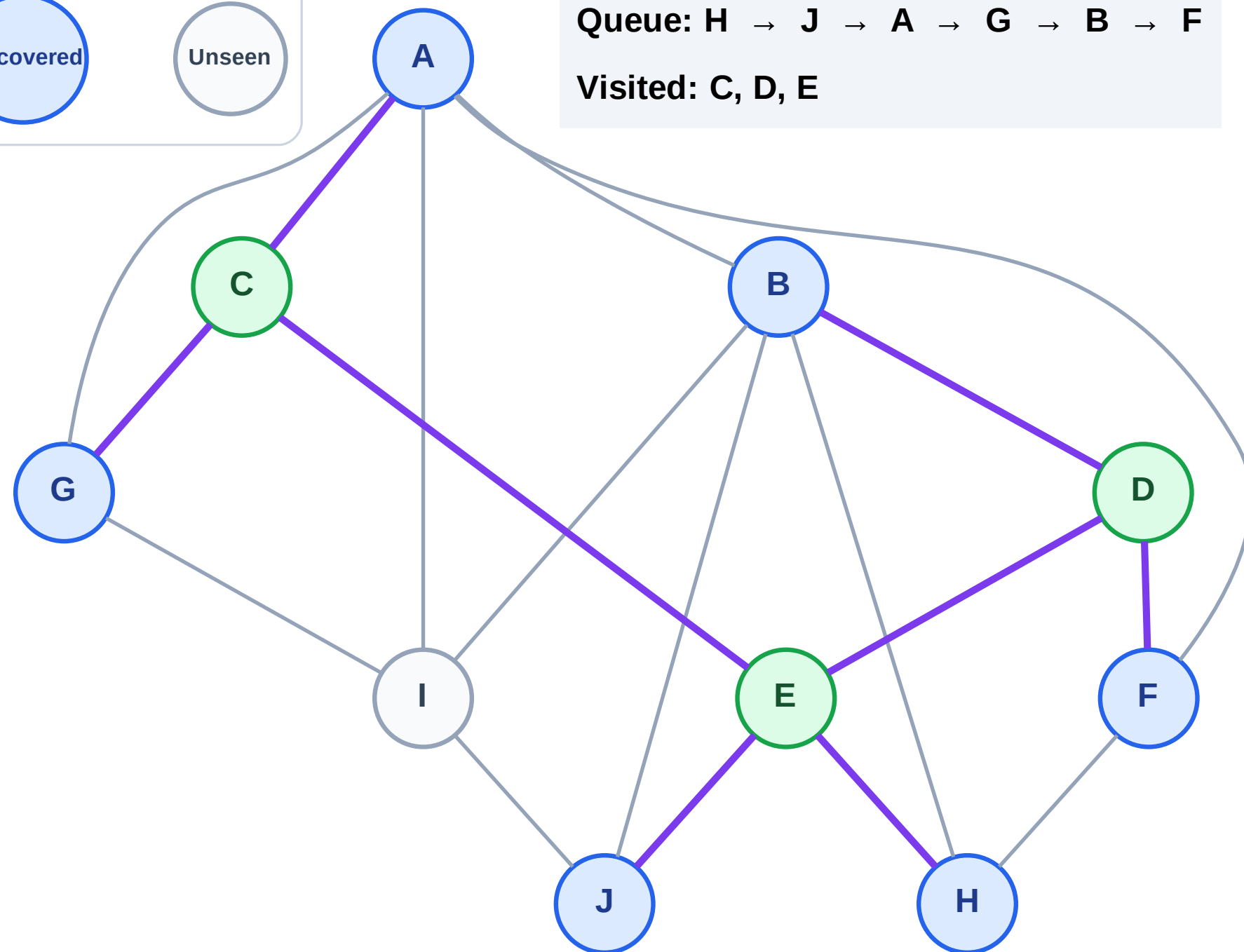
Step 7: Discover B, F from D

Legend



Queue: H → J → A → G → B → F

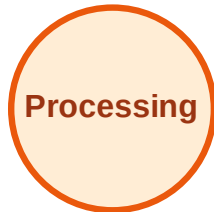
Visited: C, D, E



BFS Traversal

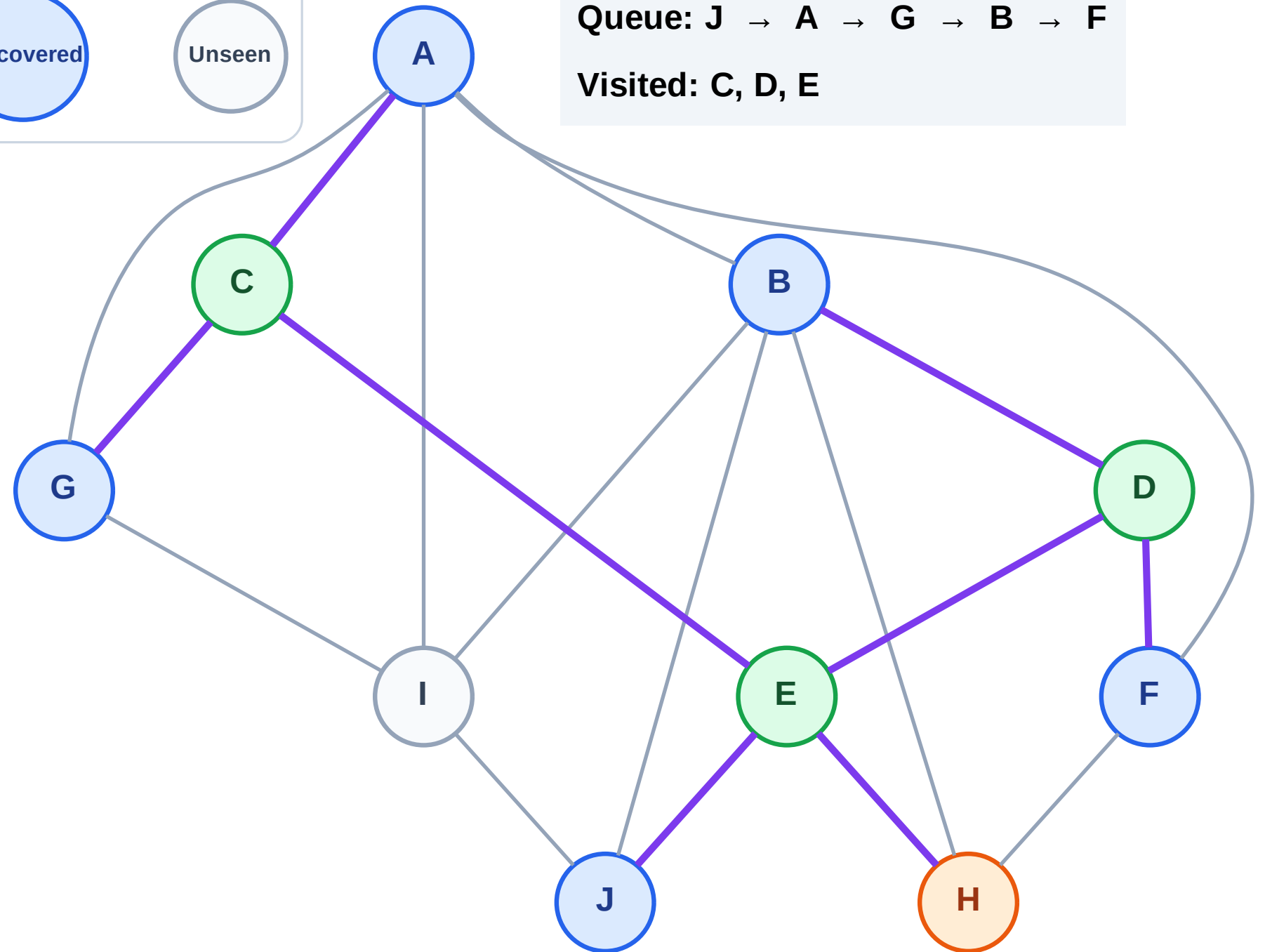
Step 8: Process node H

Legend



Queue: J → A → G → B → F

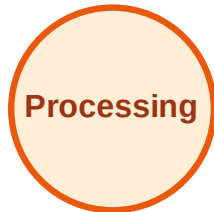
Visited: C, D, E



BFS Traversal

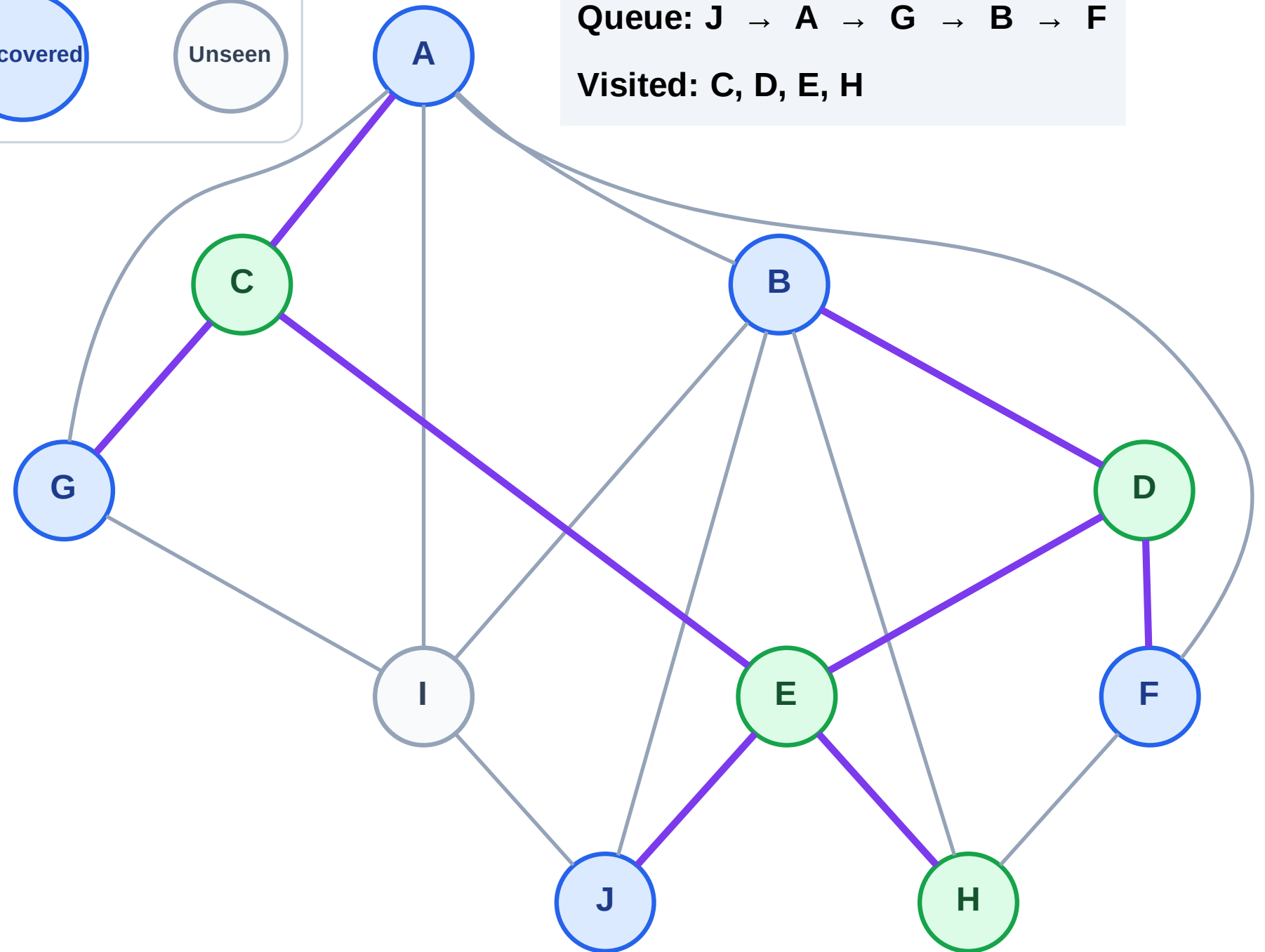
Step 9: No new neighbors from H

Legend



Queue: J → A → G → B → F

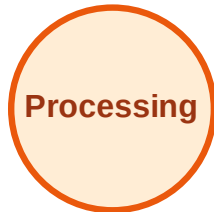
Visited: C, D, E, H



BFS Traversal

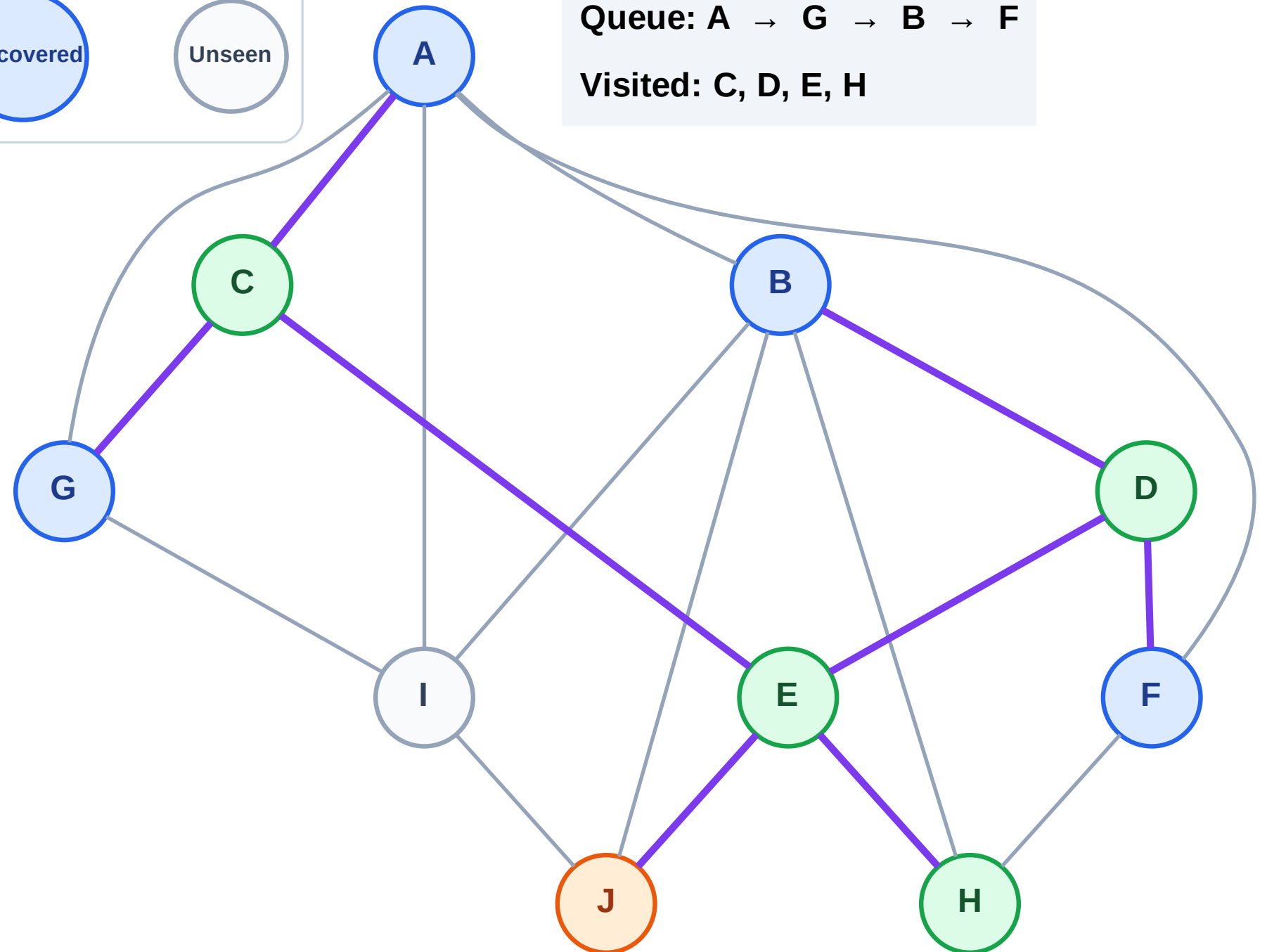
Step 10: Process node J

Legend



Queue: A → G → B → F

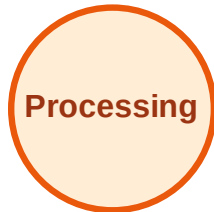
Visited: C, D, E, H



BFS Traversal

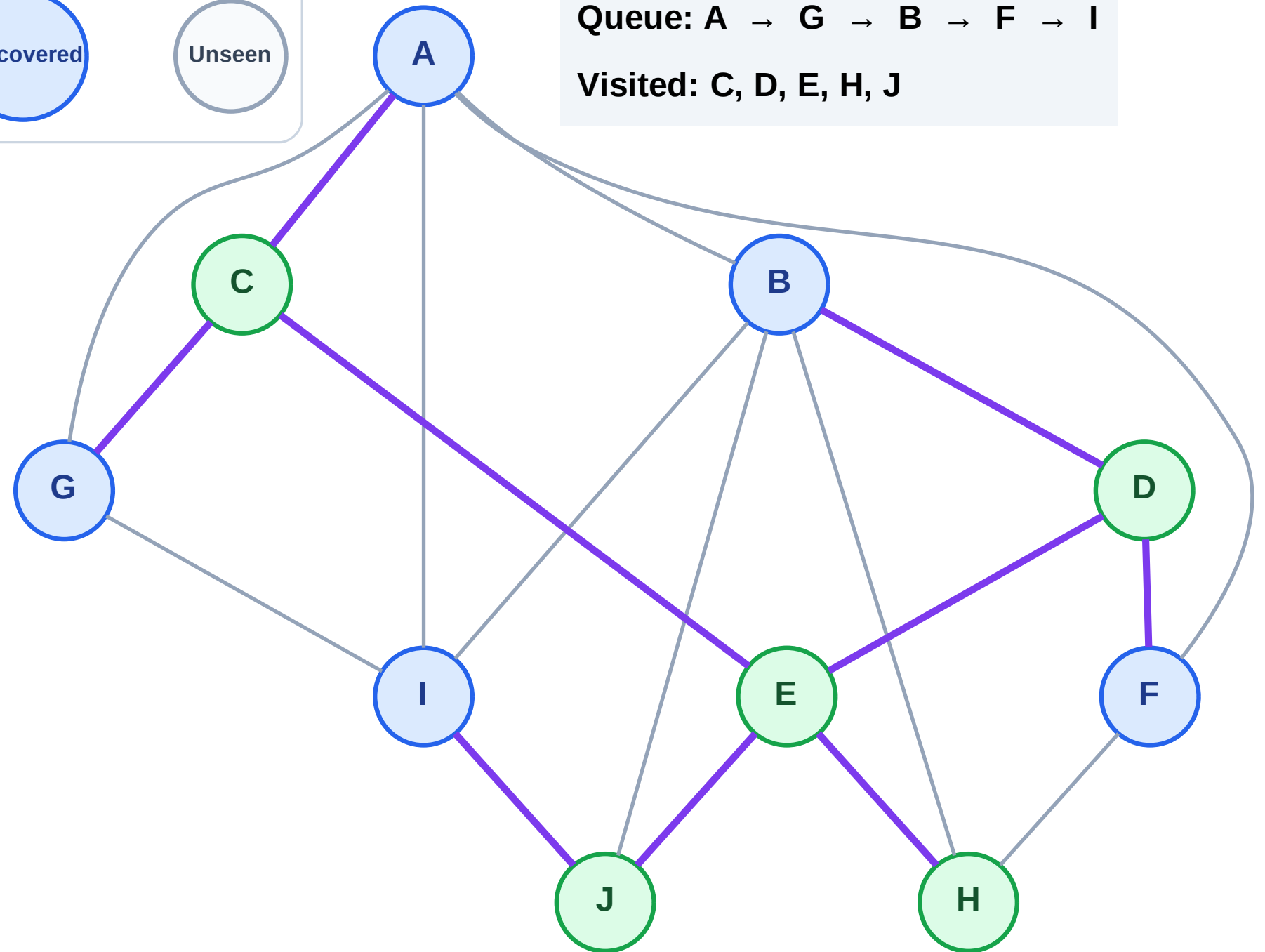
Step 11: Discover I from J

Legend



Queue: A → G → B → F → I

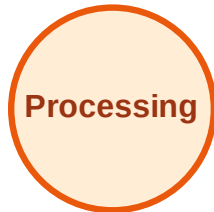
Visited: C, D, E, H, J



BFS Traversal

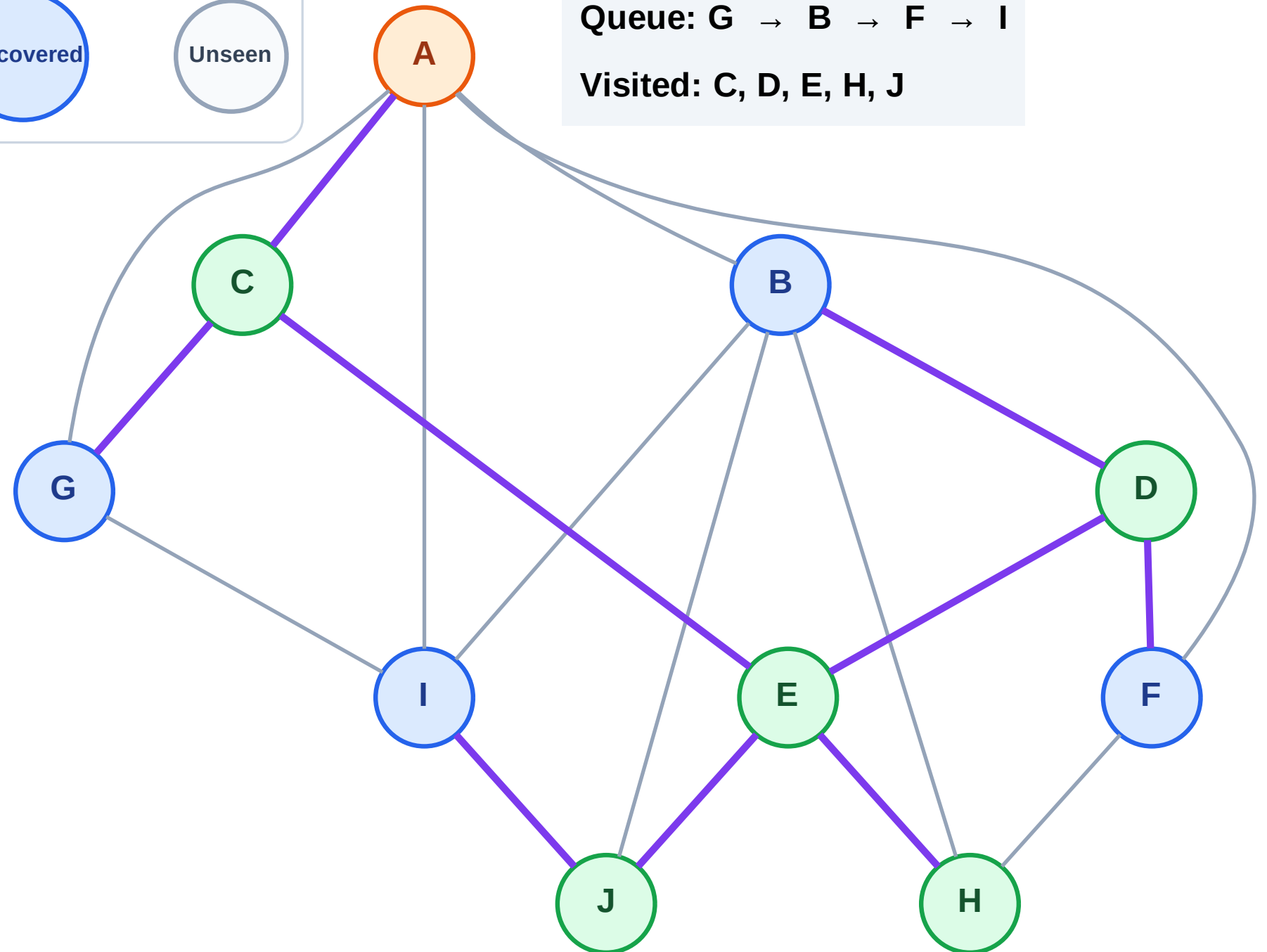
Step 12: Process node A

Legend



Queue: G → B → F → I

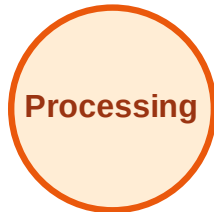
Visited: C, D, E, H, J



BFS Traversal

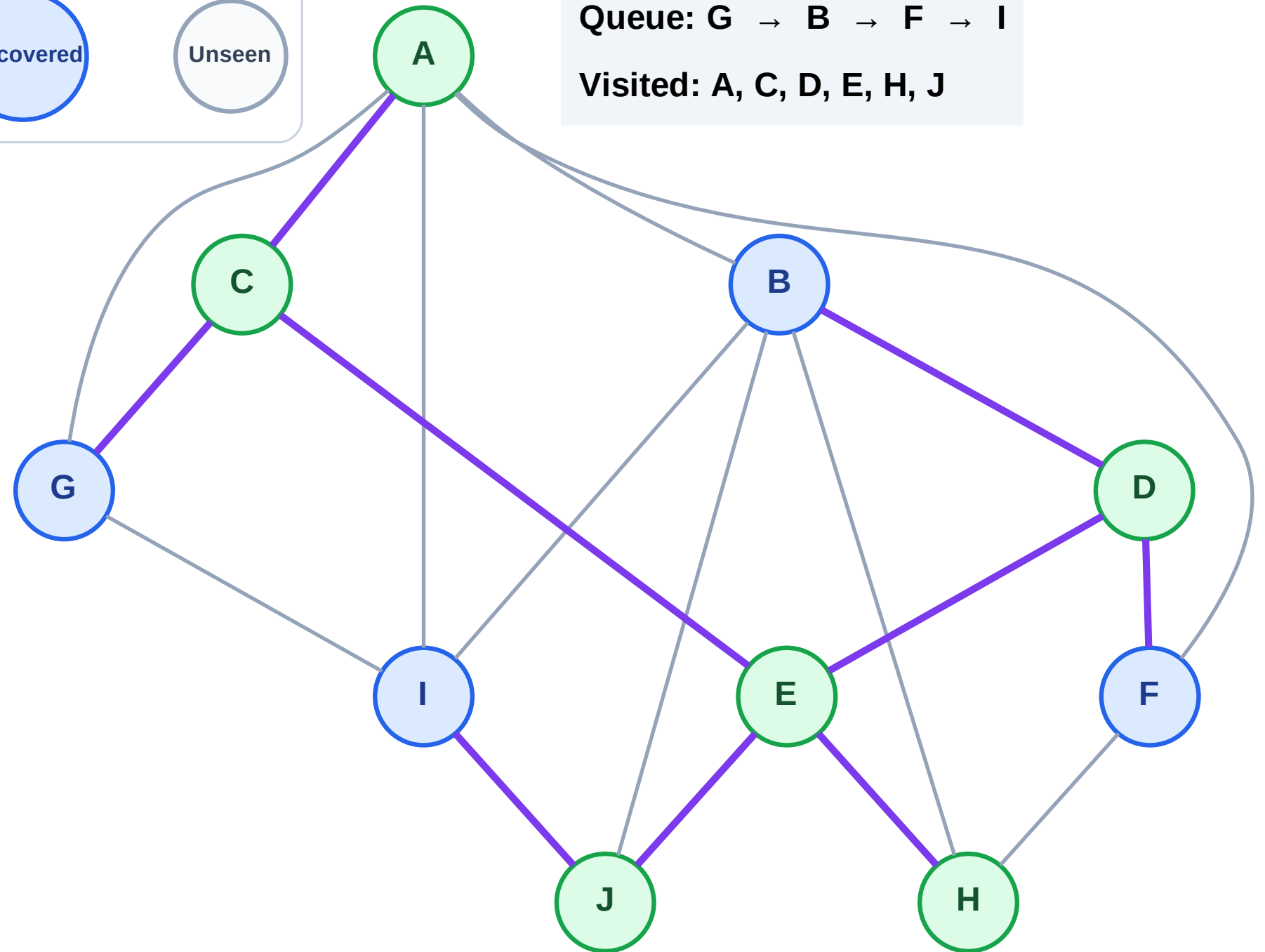
Step 13: No new neighbors from A

Legend



Queue: G → B → F → I

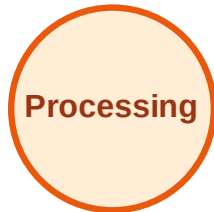
Visited: A, C, D, E, H, J



BFS Traversal

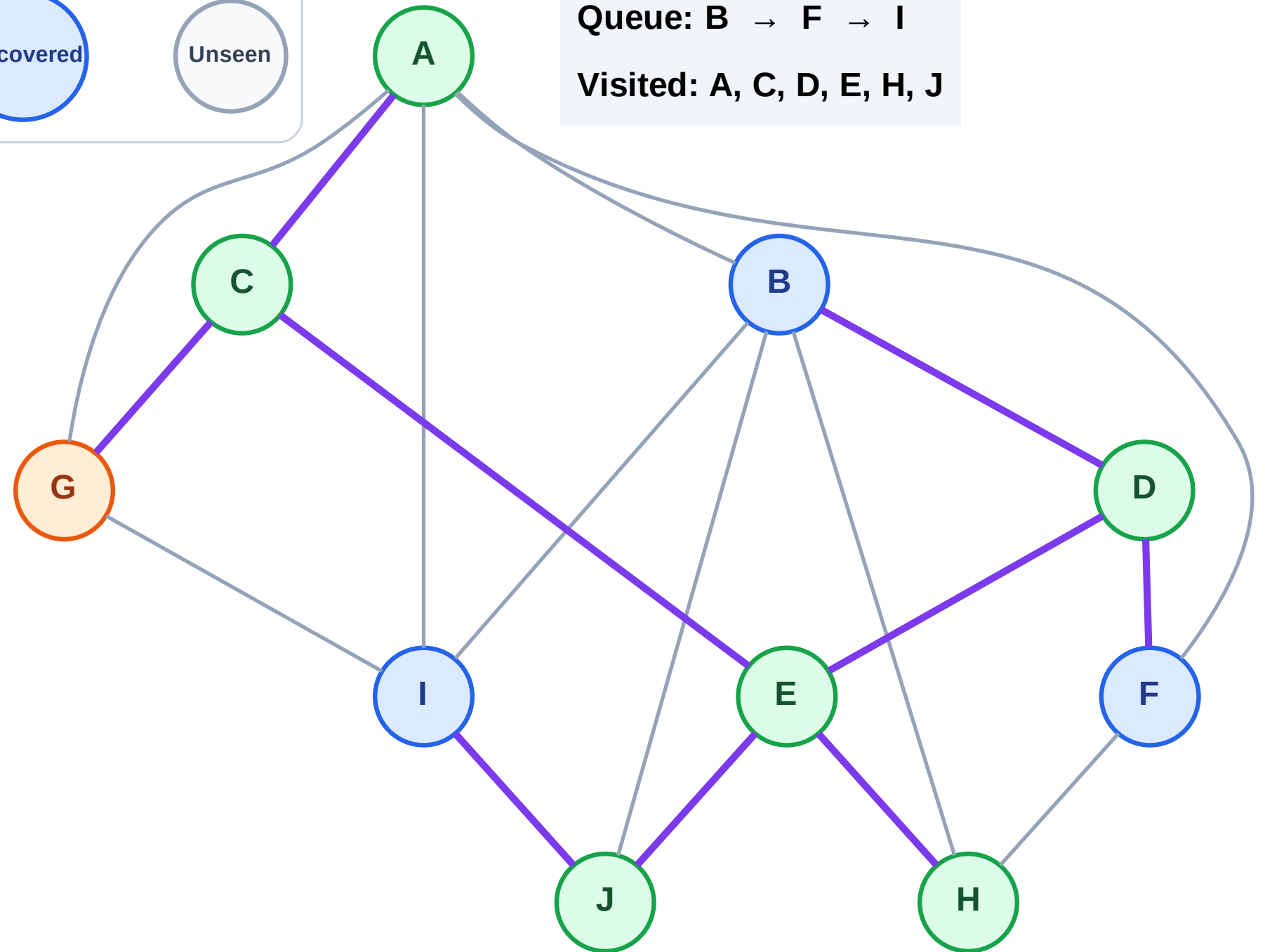
Step 14: Process node G

Legend



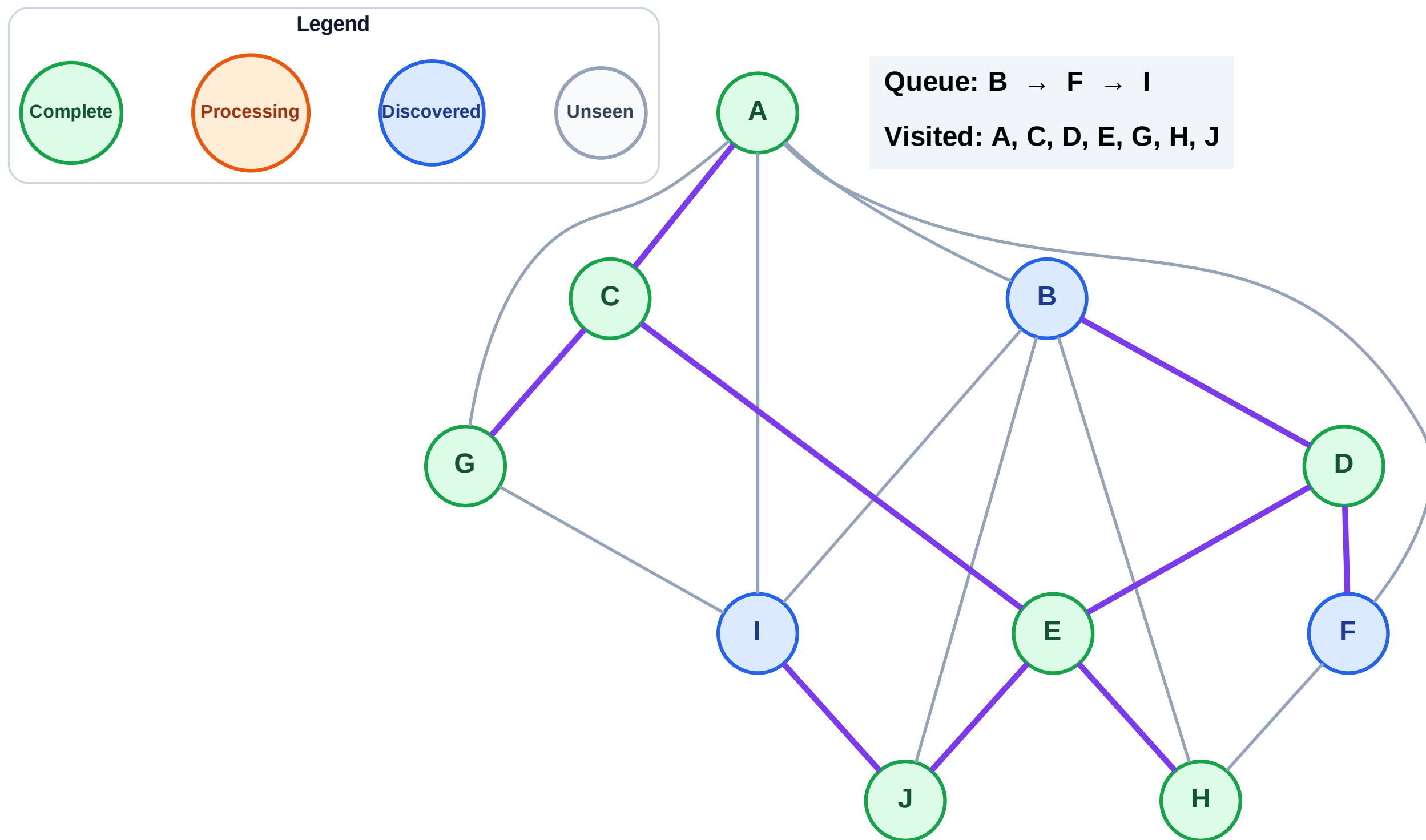
Queue: B → F → I

Visited: A, C, D, E, H, J



Step 15: No new neighbors from G

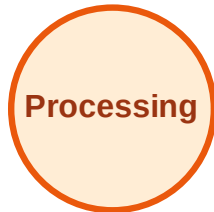
Step 15: No new neighbors from G



BFS Traversal

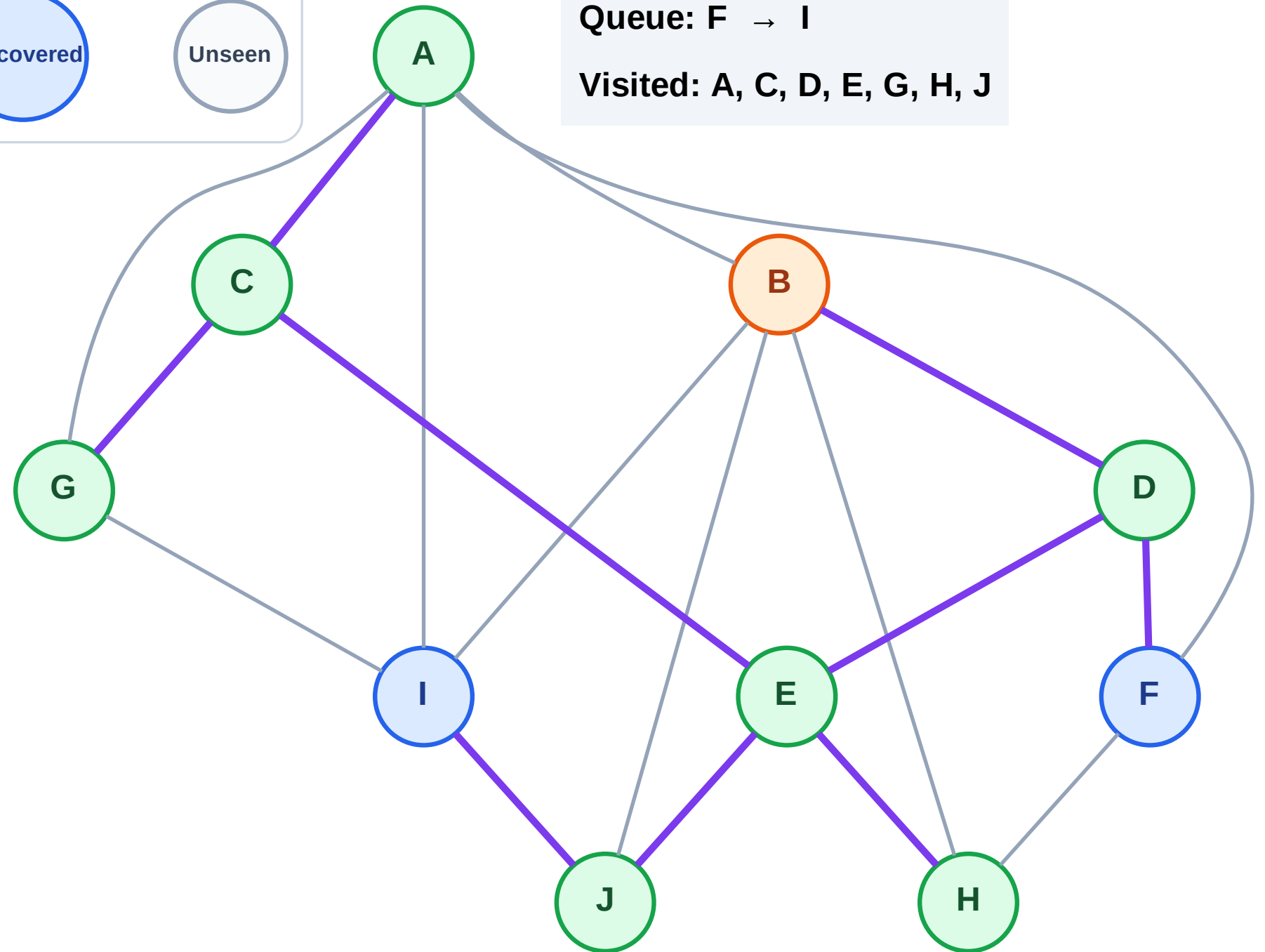
Step 16: Process node B

Legend



Queue: F → I

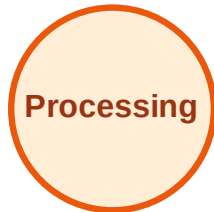
Visited: A, C, D, E, G, H, J



BFS Traversal

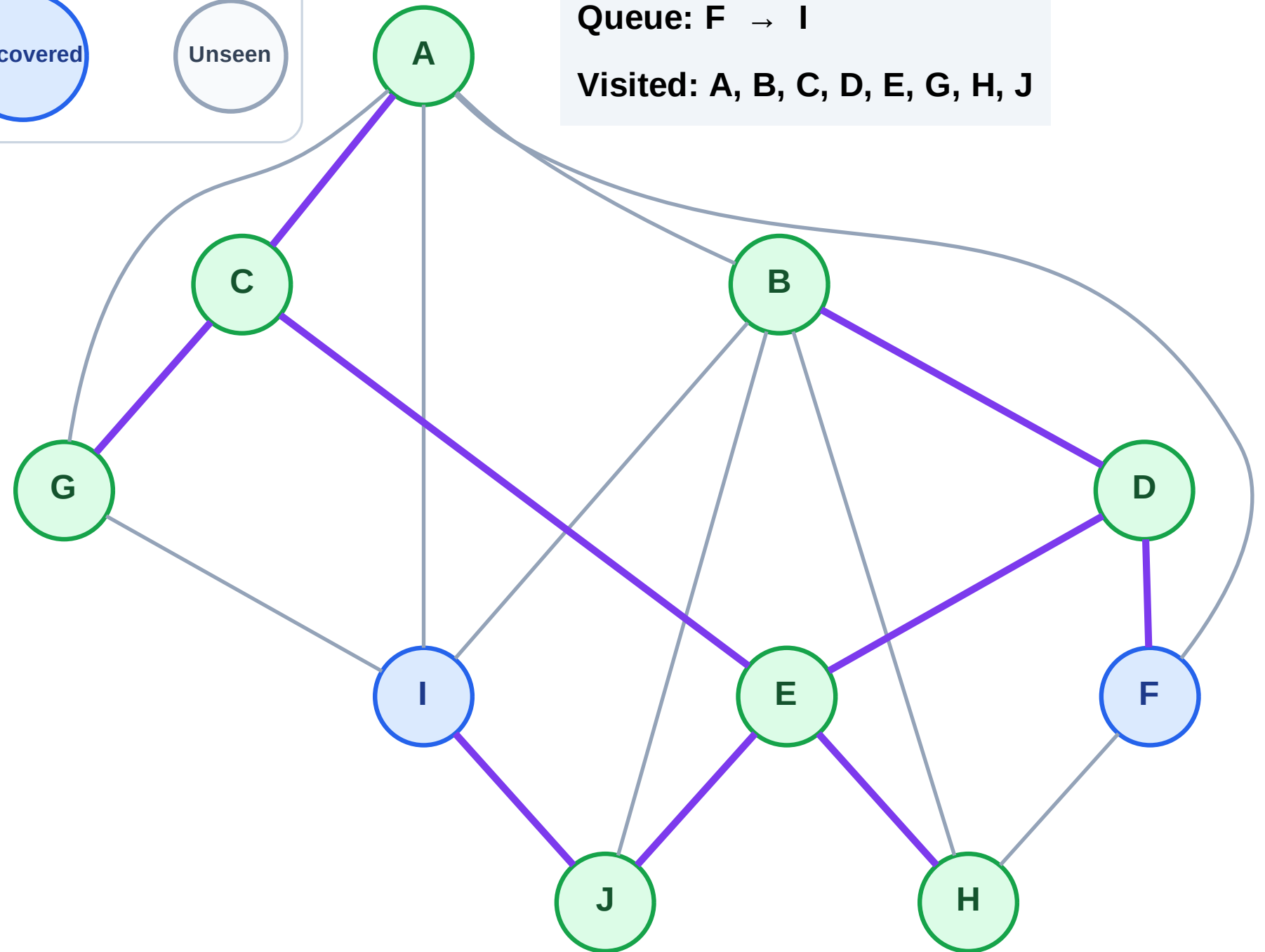
Step 17: No new neighbors from B

Legend



Queue: F → I

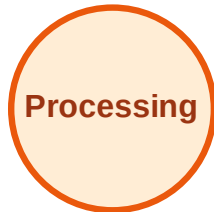
Visited: A, B, C, D, E, G, H, J



BFS Traversal

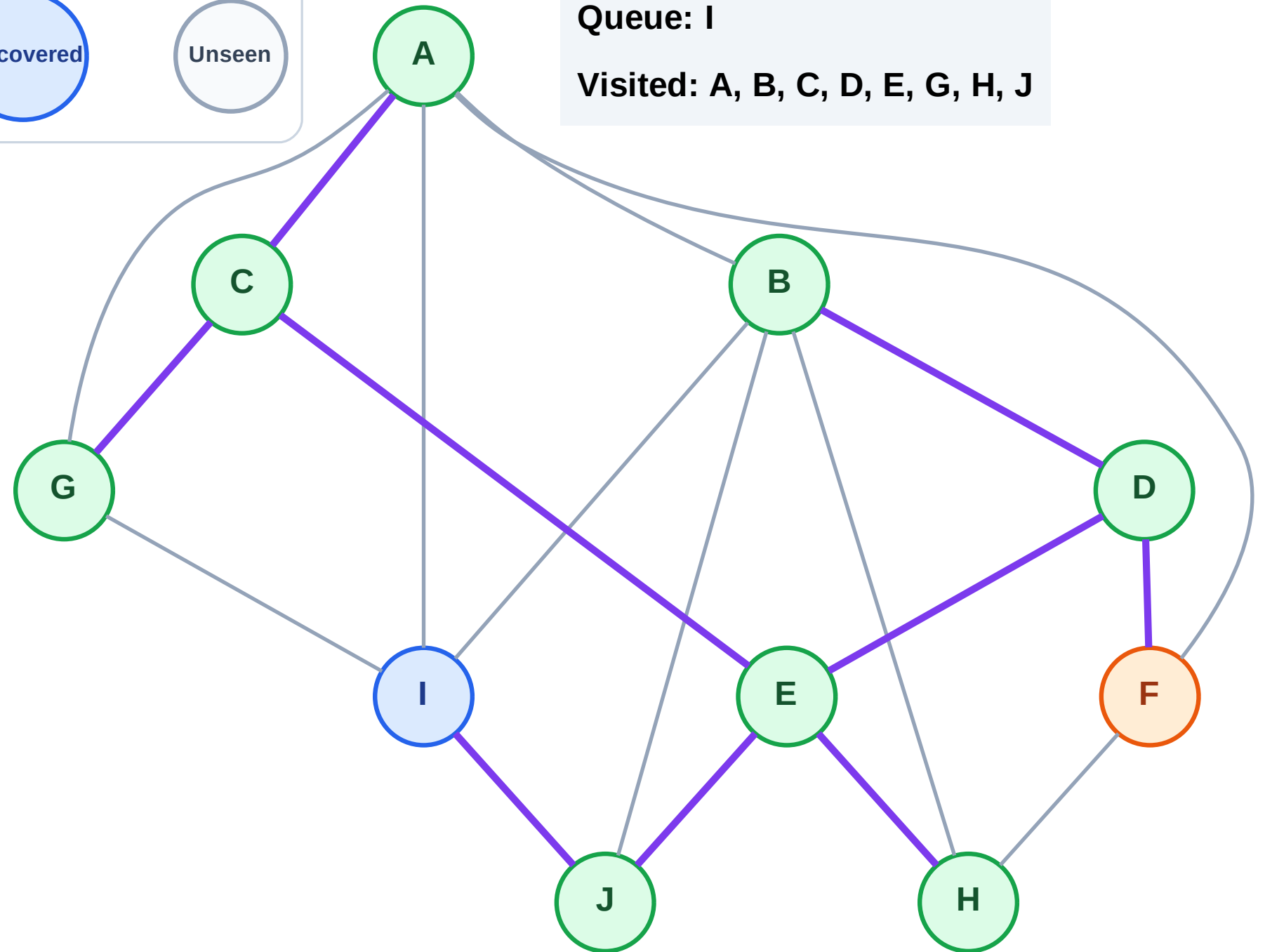
Step 18: Process node F

Legend



Queue: I

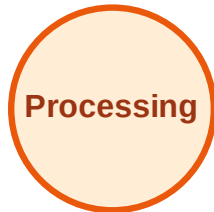
Visited: A, B, C, D, E, G, H, J



BFS Traversal

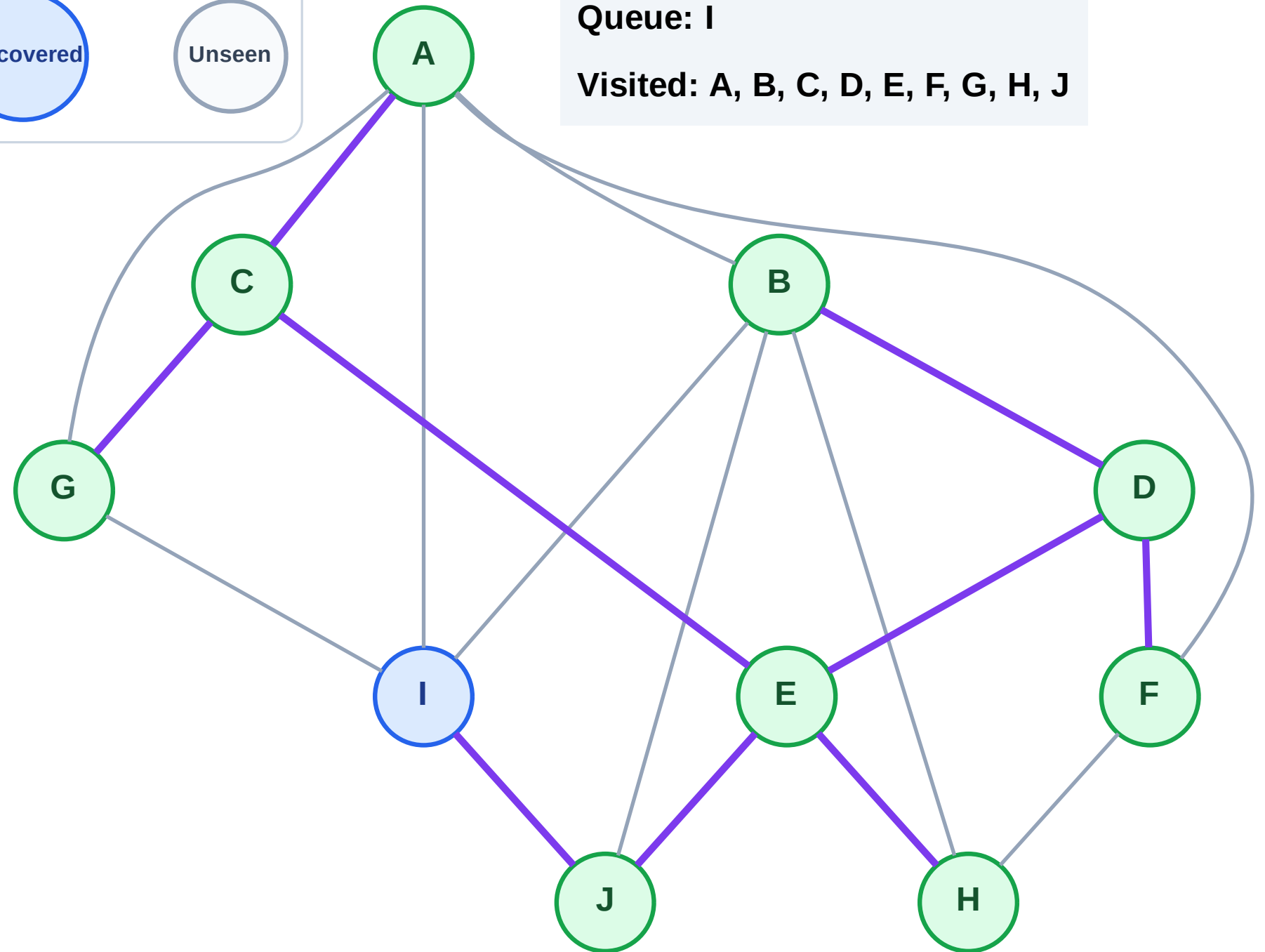
Step 19: No new neighbors from F

Legend



Queue: I

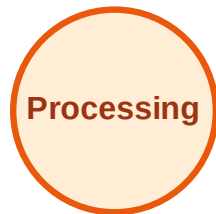
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

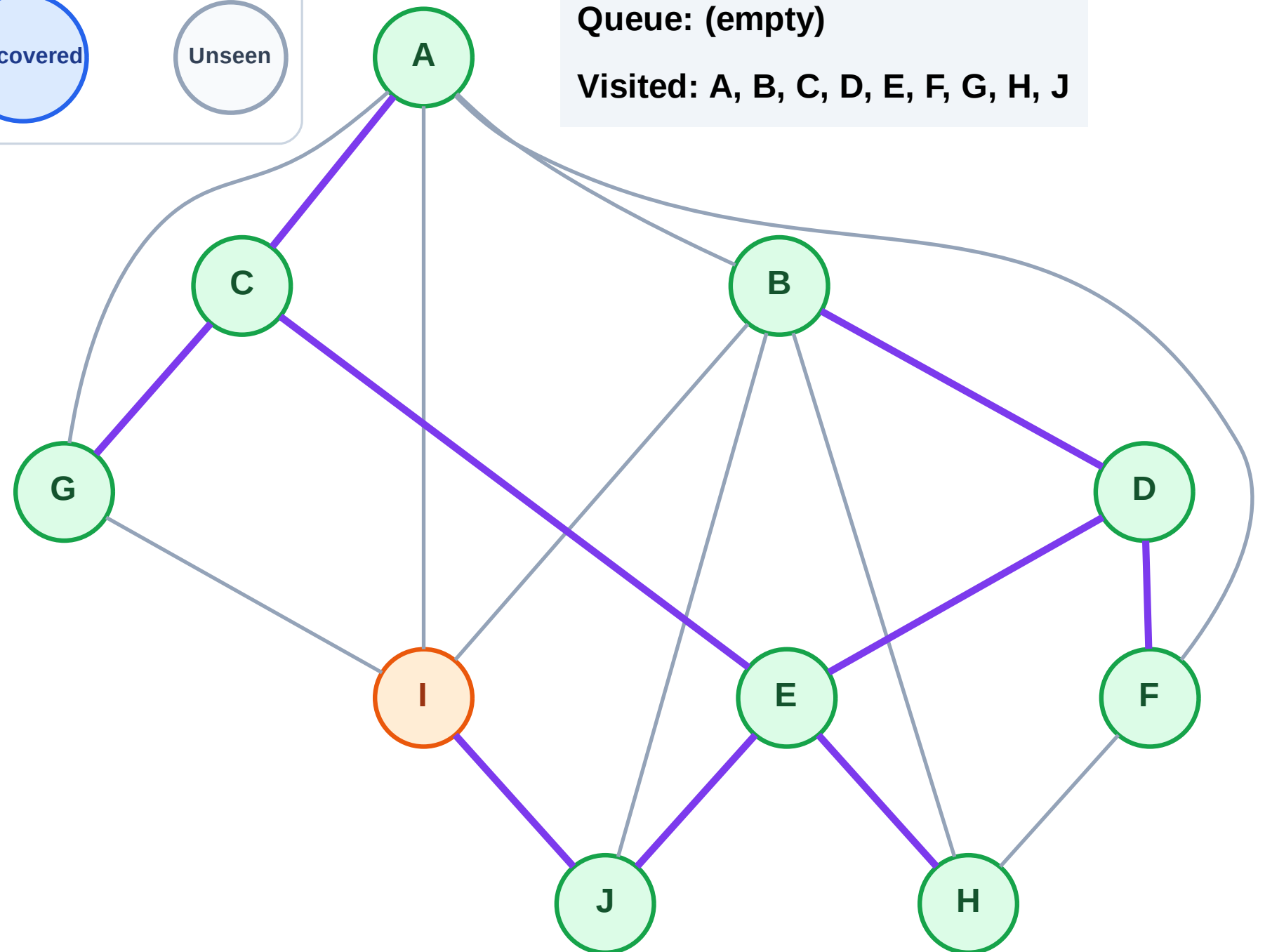
Step 20: Process node I

Legend



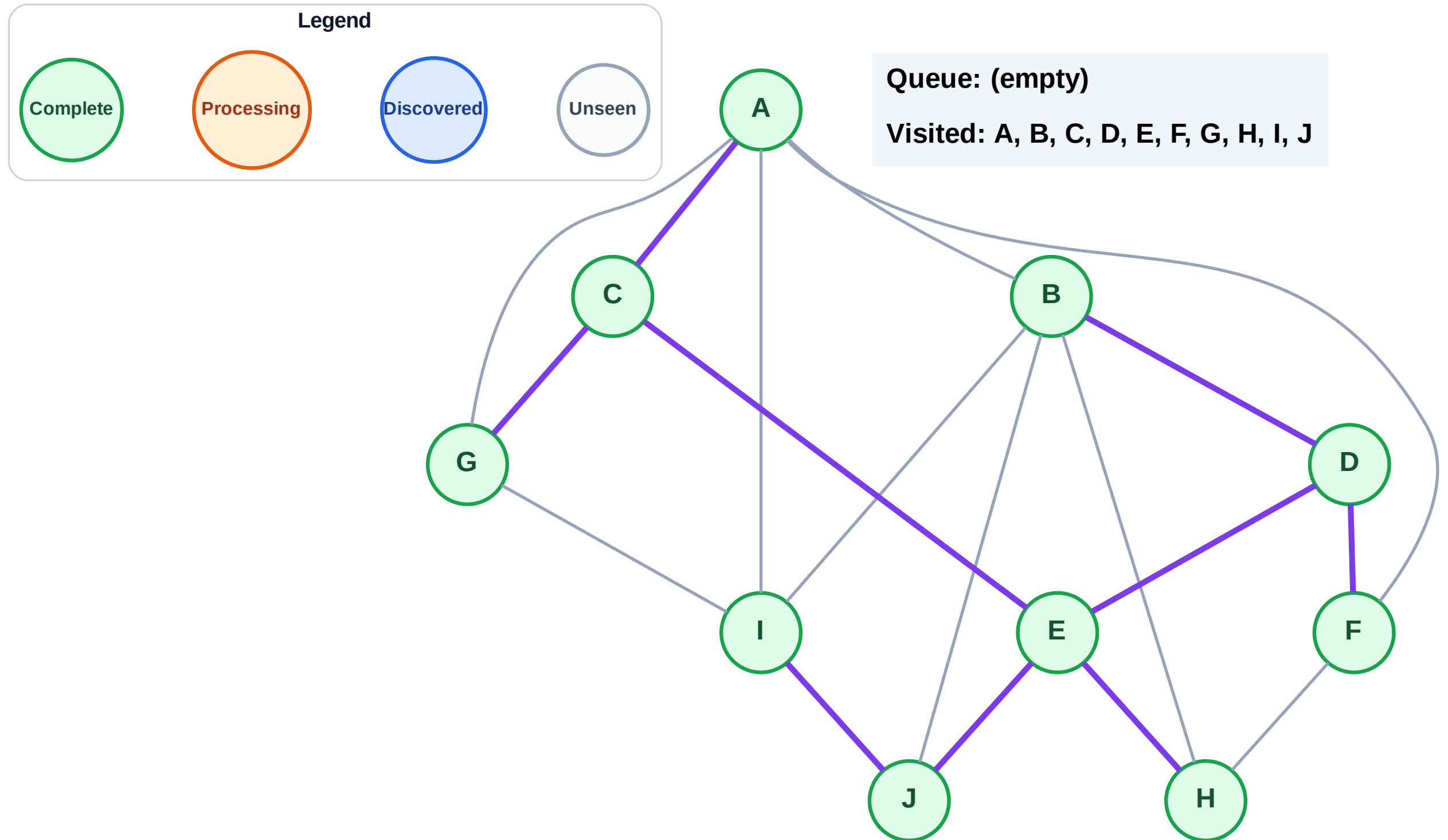
Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



Step 21: No new neighbors from I

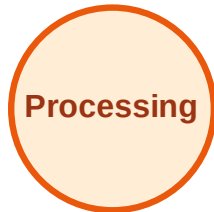
Step 21: No new neighbors from I



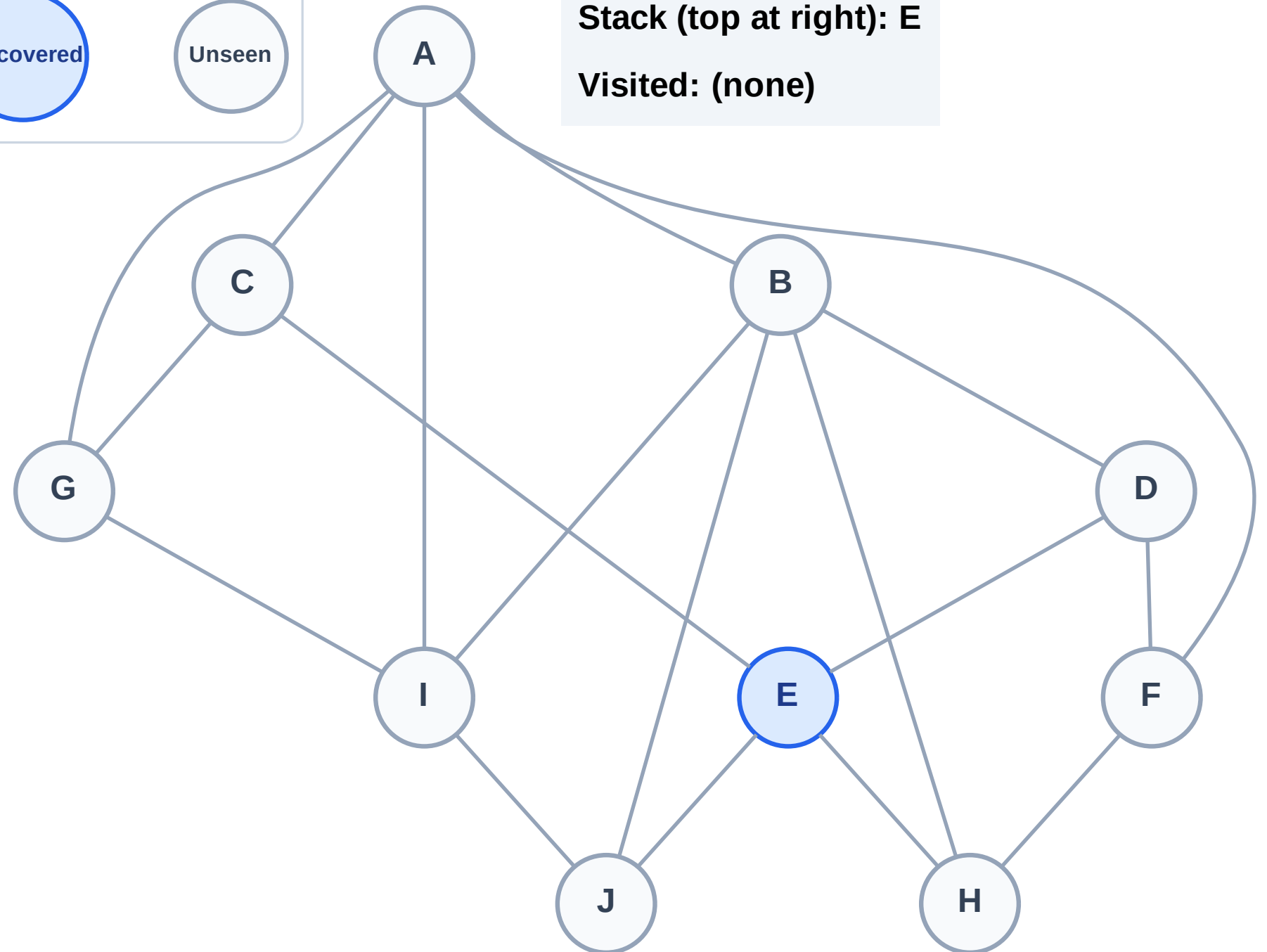
DFS Traversal

Step 1: Start at E

Legend



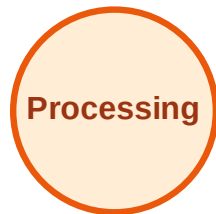
Stack (top at right): E
Visited: (none)



DFS Traversal

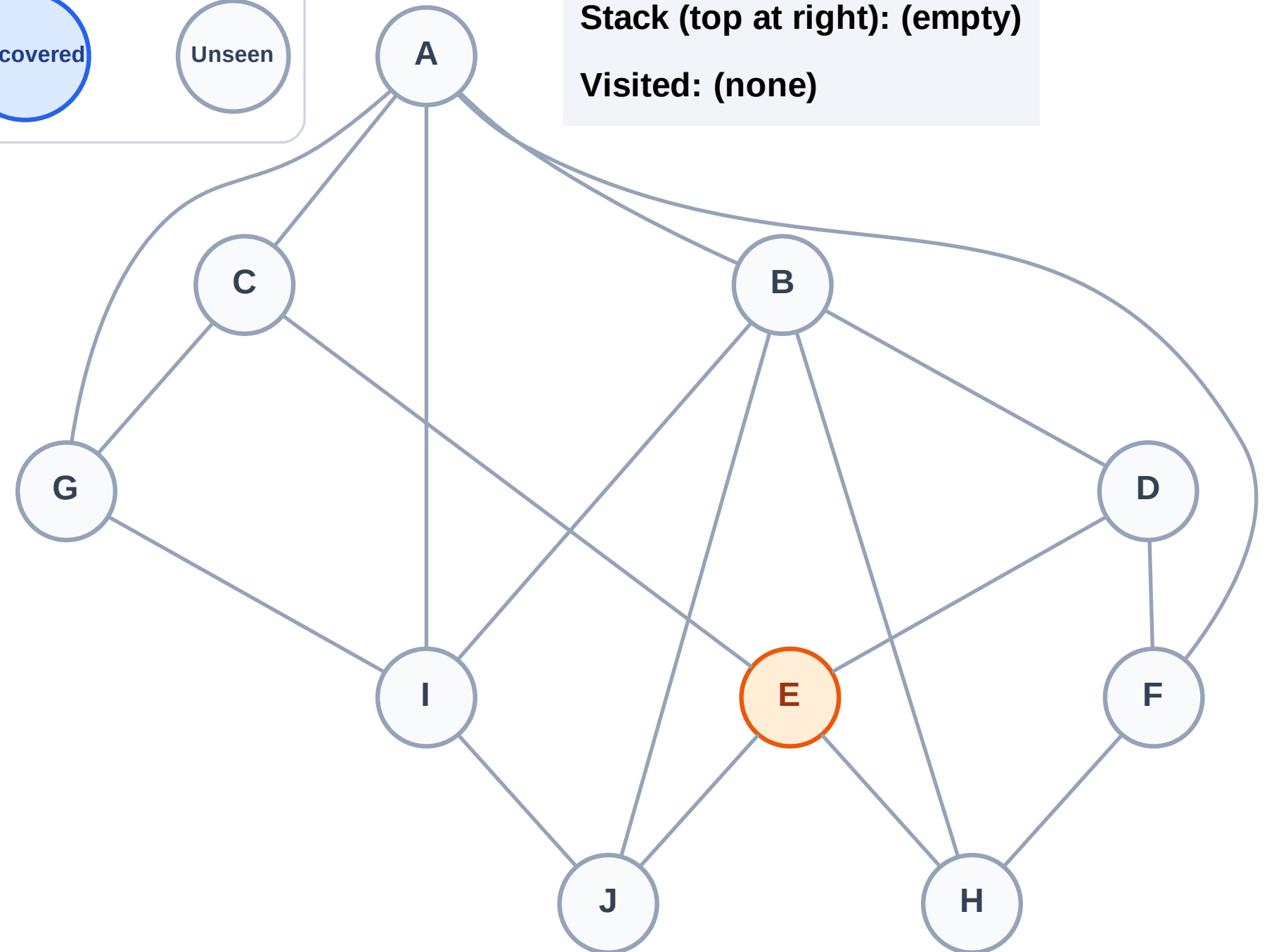
Step 2: Process node E

Legend



Stack (top at right): (empty)

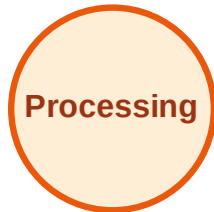
Visited: (none)



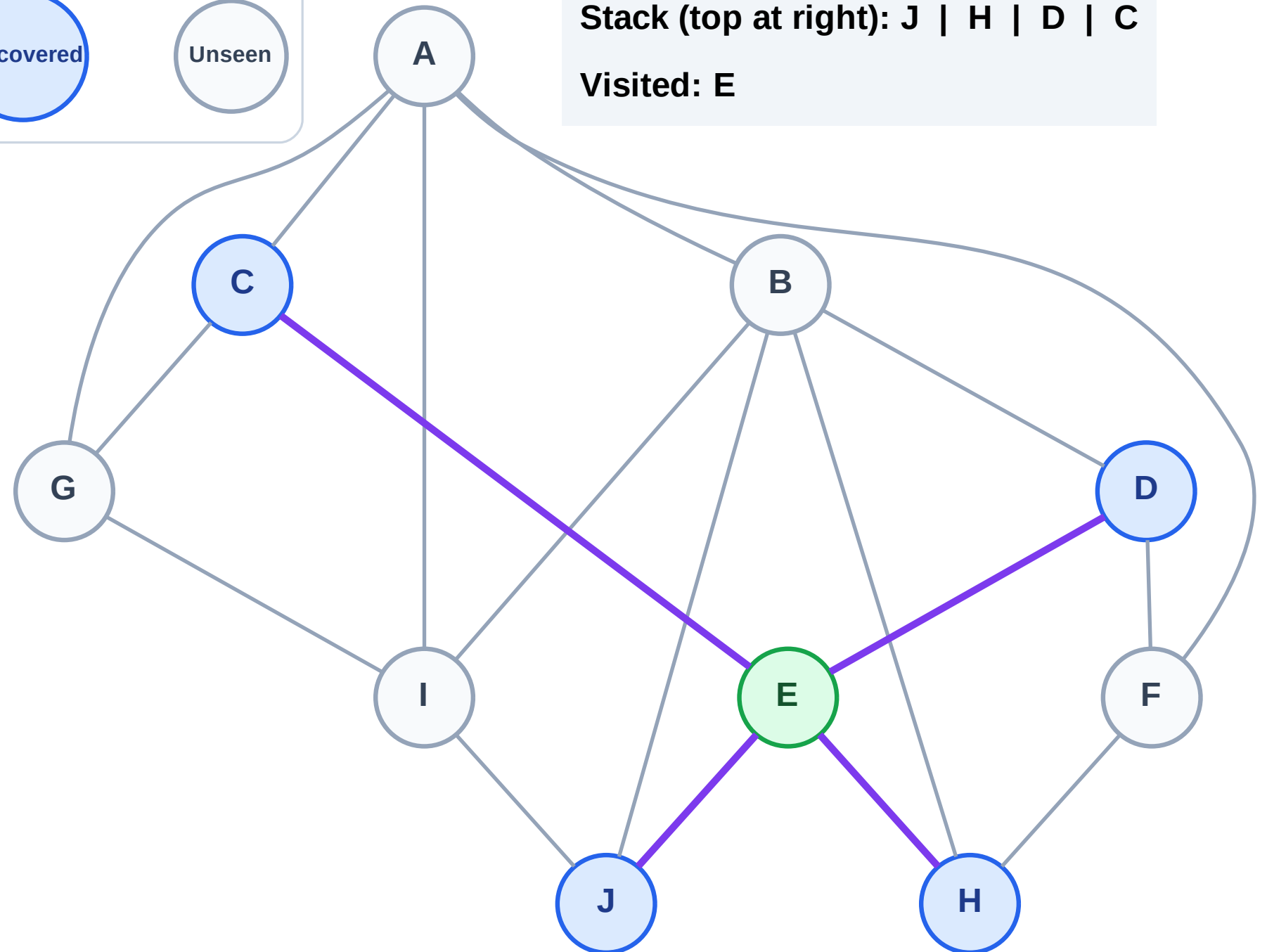
DFS Traversal

Step 3: Discover J, H, D, C from E

Legend



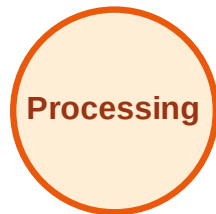
Stack (top at right): J | H | D | C
Visited: E



DFS Traversal

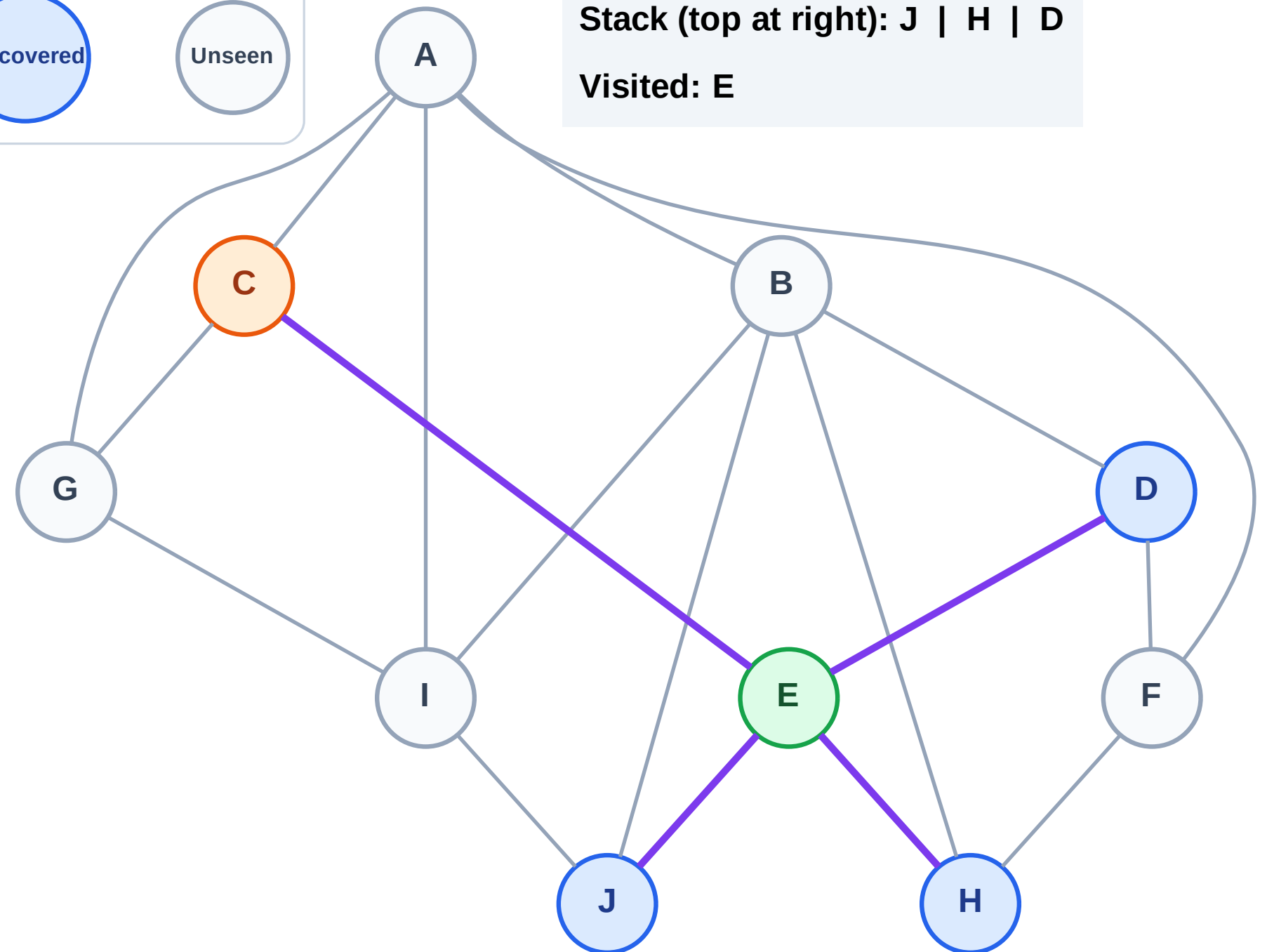
Step 4: Process node C

Legend



Stack (top at right): J | H | D

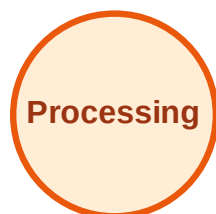
Visited: E



DFS Traversal

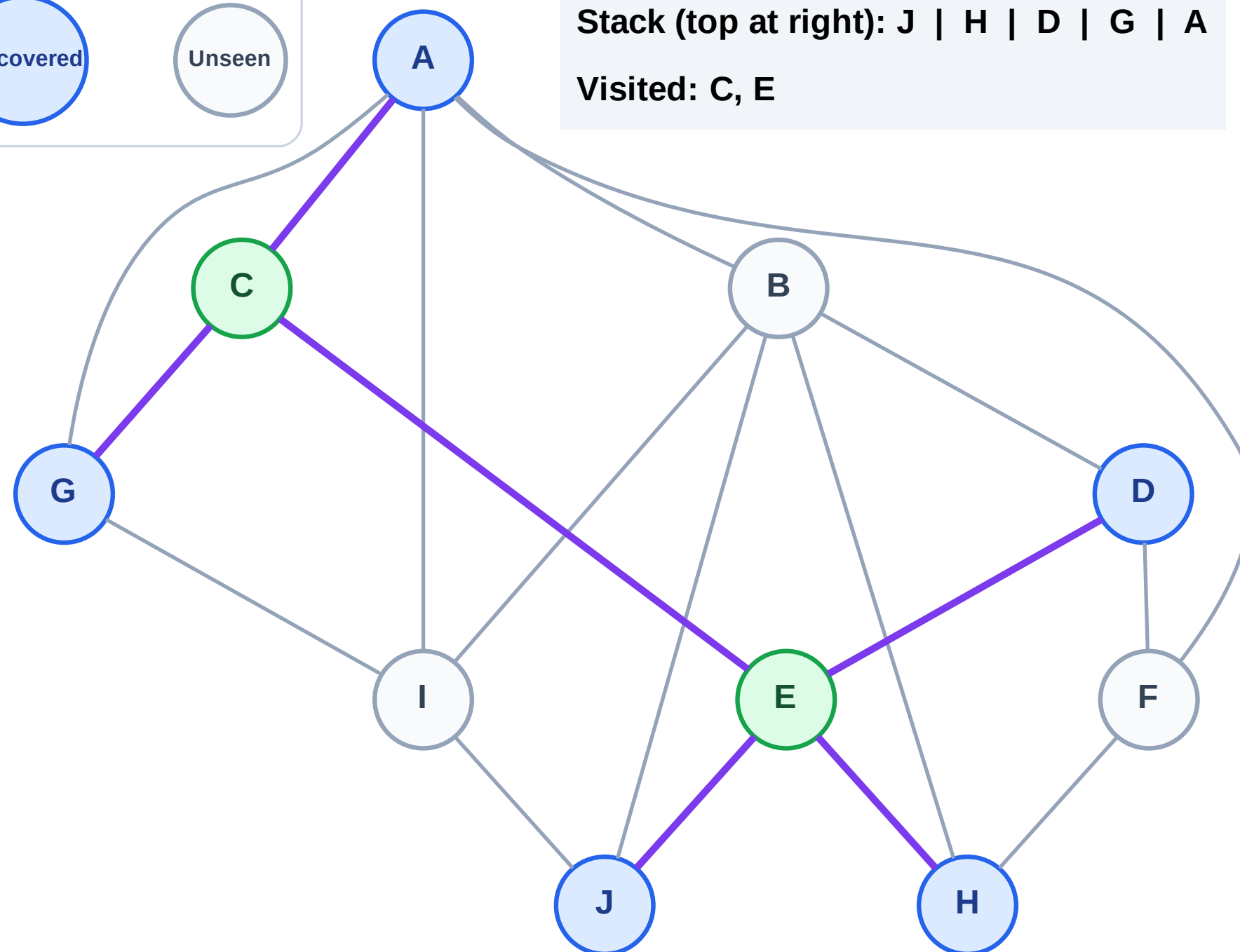
Step 5: Discover G, A from C

Legend



Stack (top at right): J | H | D | G | A

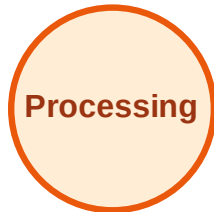
Visited: C, E



DFS Traversal

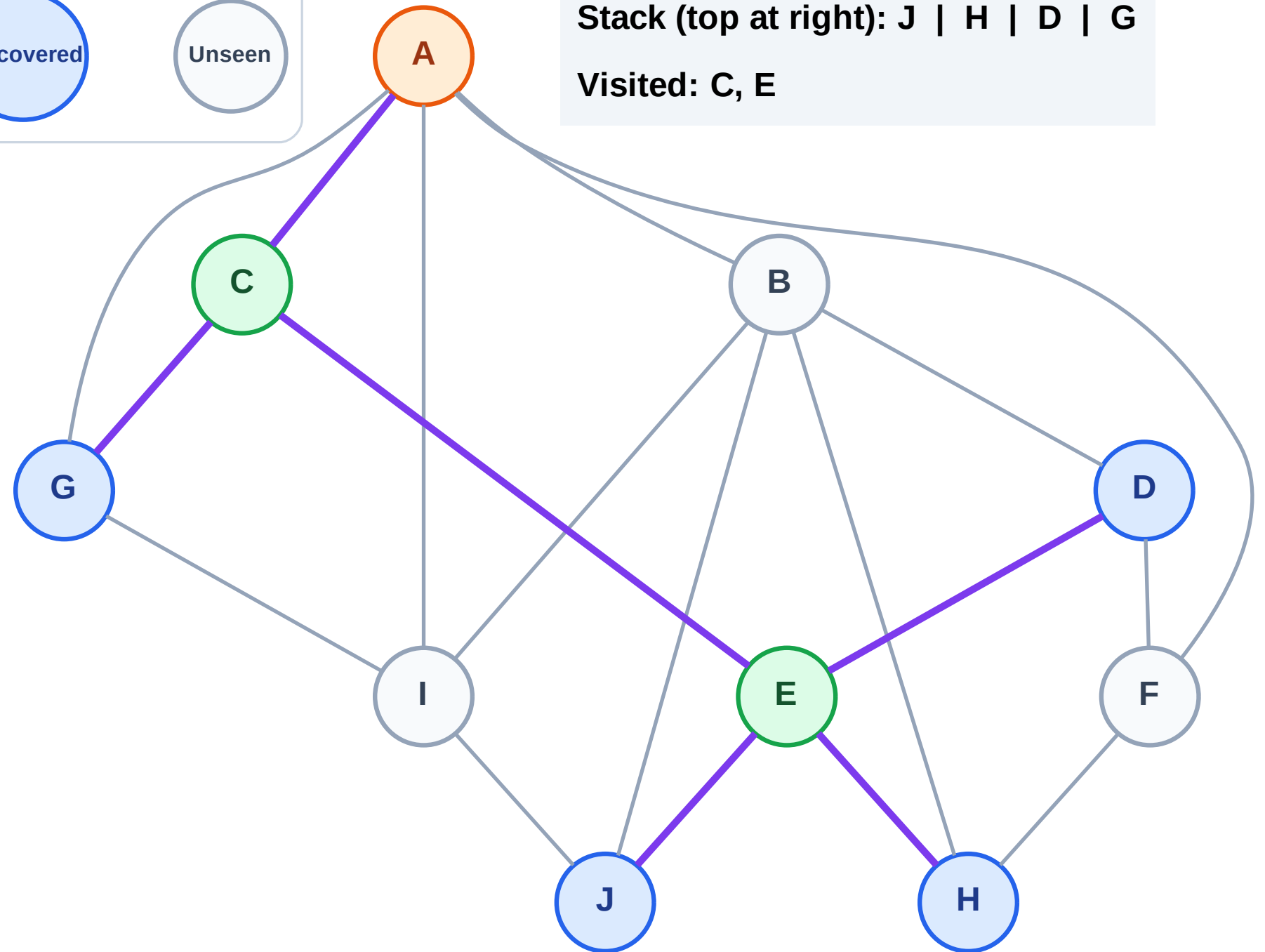
Step 6: Process node A

Legend



Stack (top at right): J | H | D | G

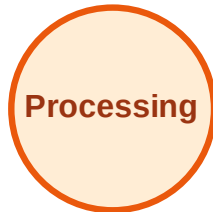
Visited: C, E



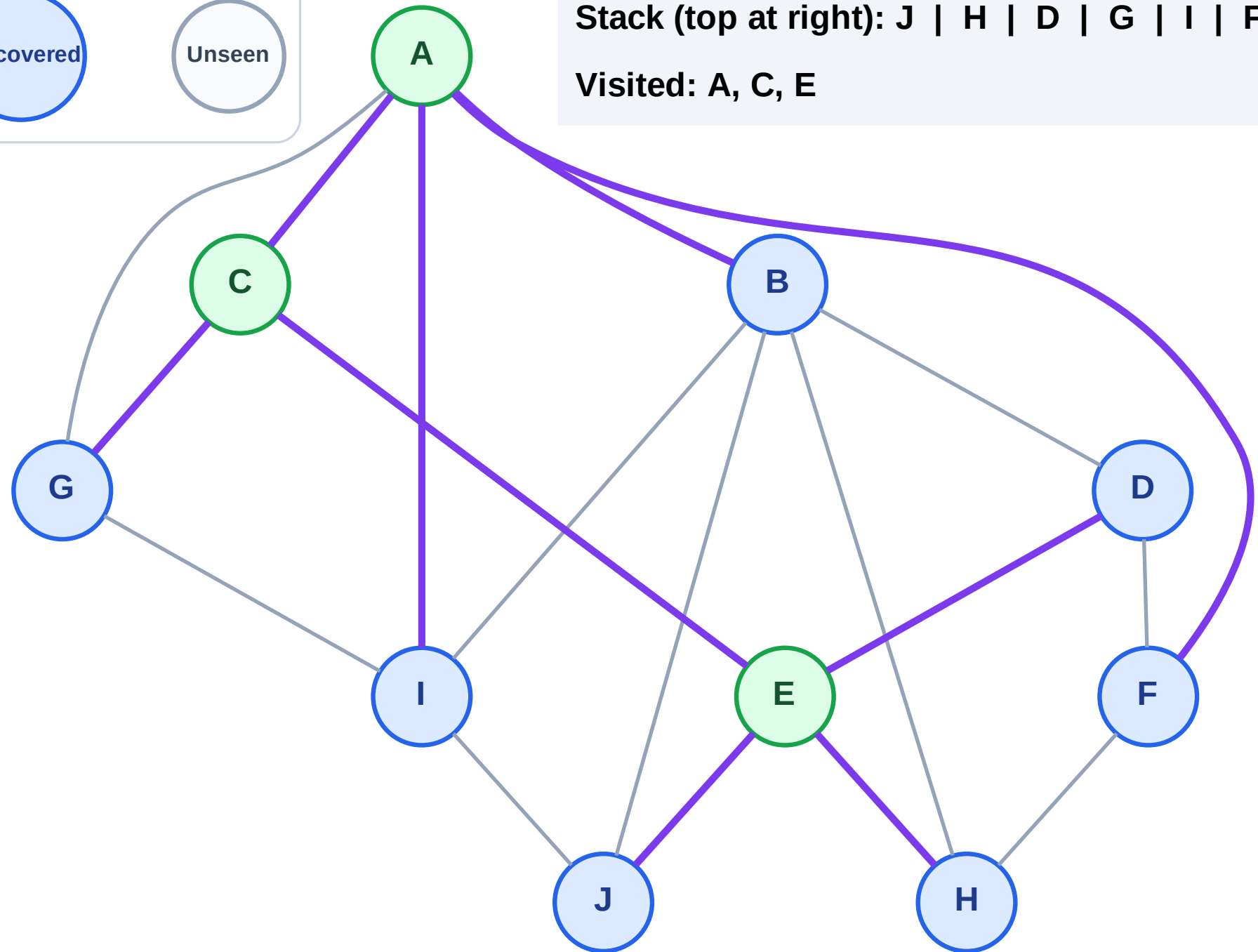
DFS Traversal

Step 7: Discover I, F, B from A

Legend



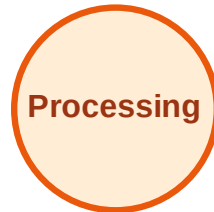
Stack (top at right): J | H | D | G | I | F | B
Visited: A, C, E



DFS Traversal

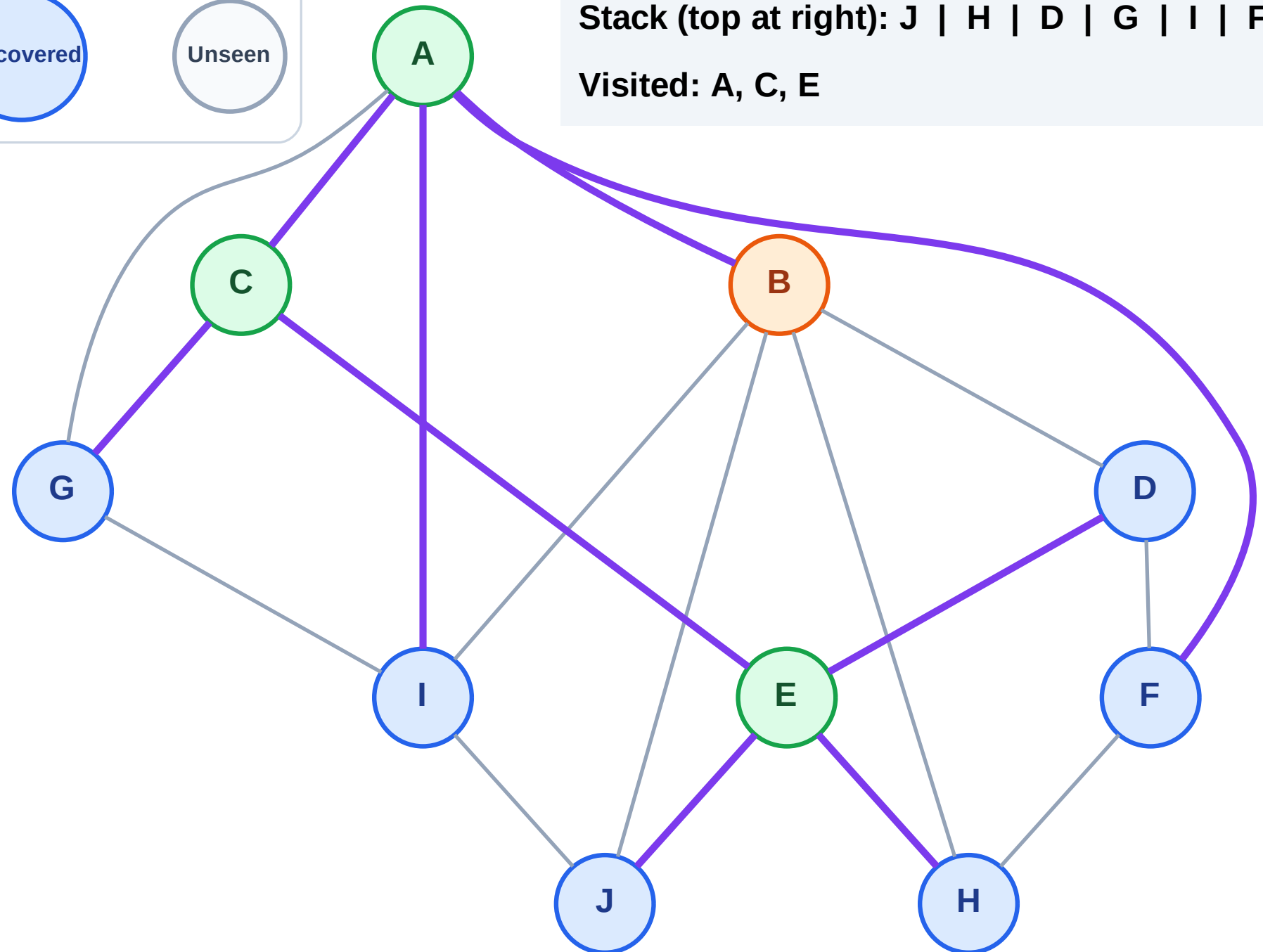
Step 8: Process node B

Legend



Stack (top at right): J | H | D | G | I | F

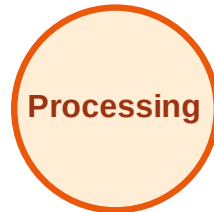
Visited: A, C, E



DFS Traversal

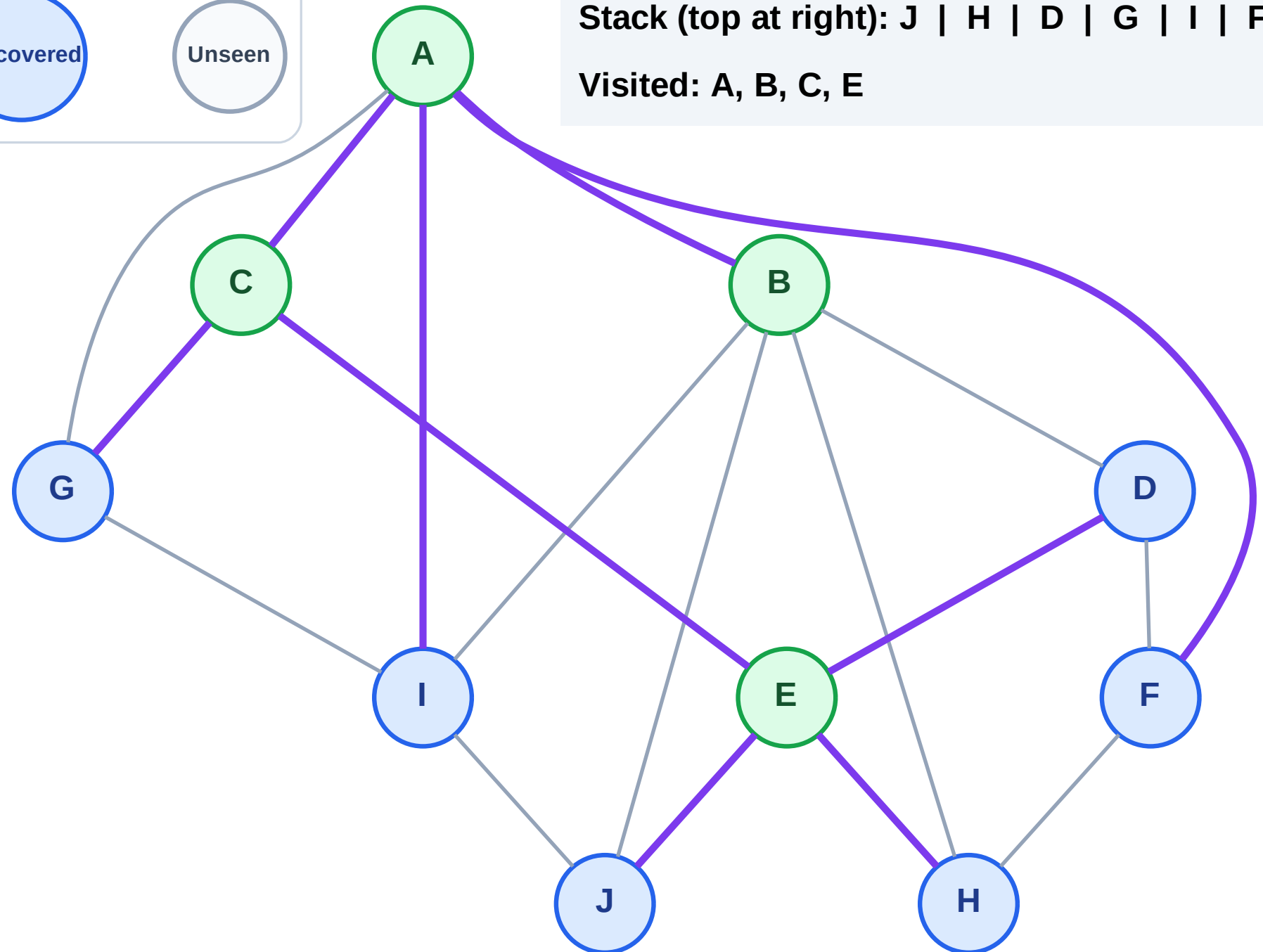
Step 9: No new neighbors from B

Legend



Stack (top at right): J | H | D | G | I | F

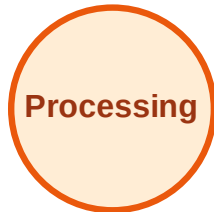
Visited: A, B, C, E



DFS Traversal

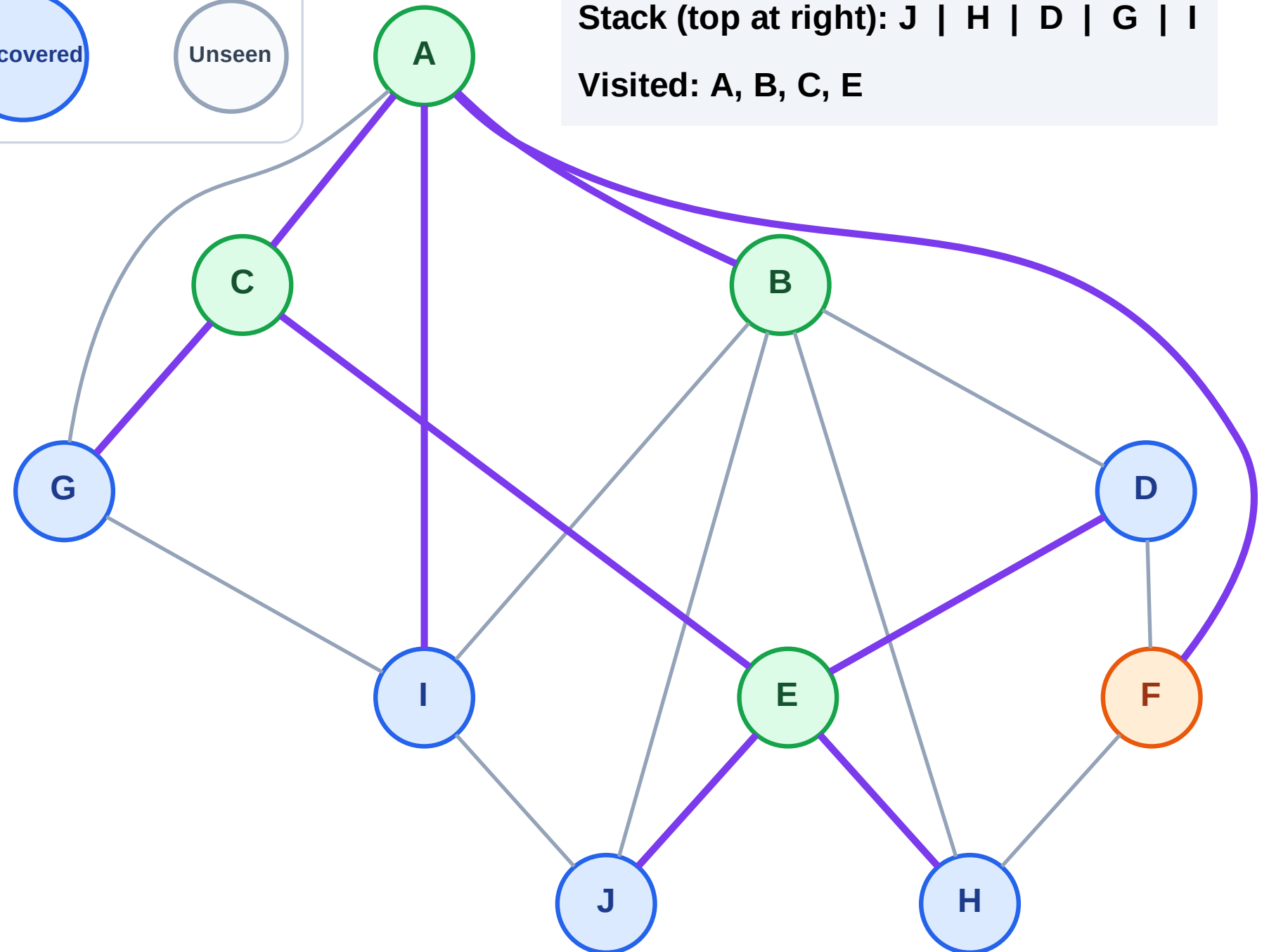
Step 10: Process node F

Legend



Stack (top at right): J | H | D | G | I

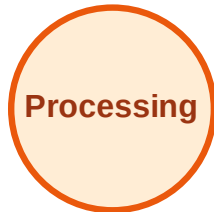
Visited: A, B, C, E



DFS Traversal

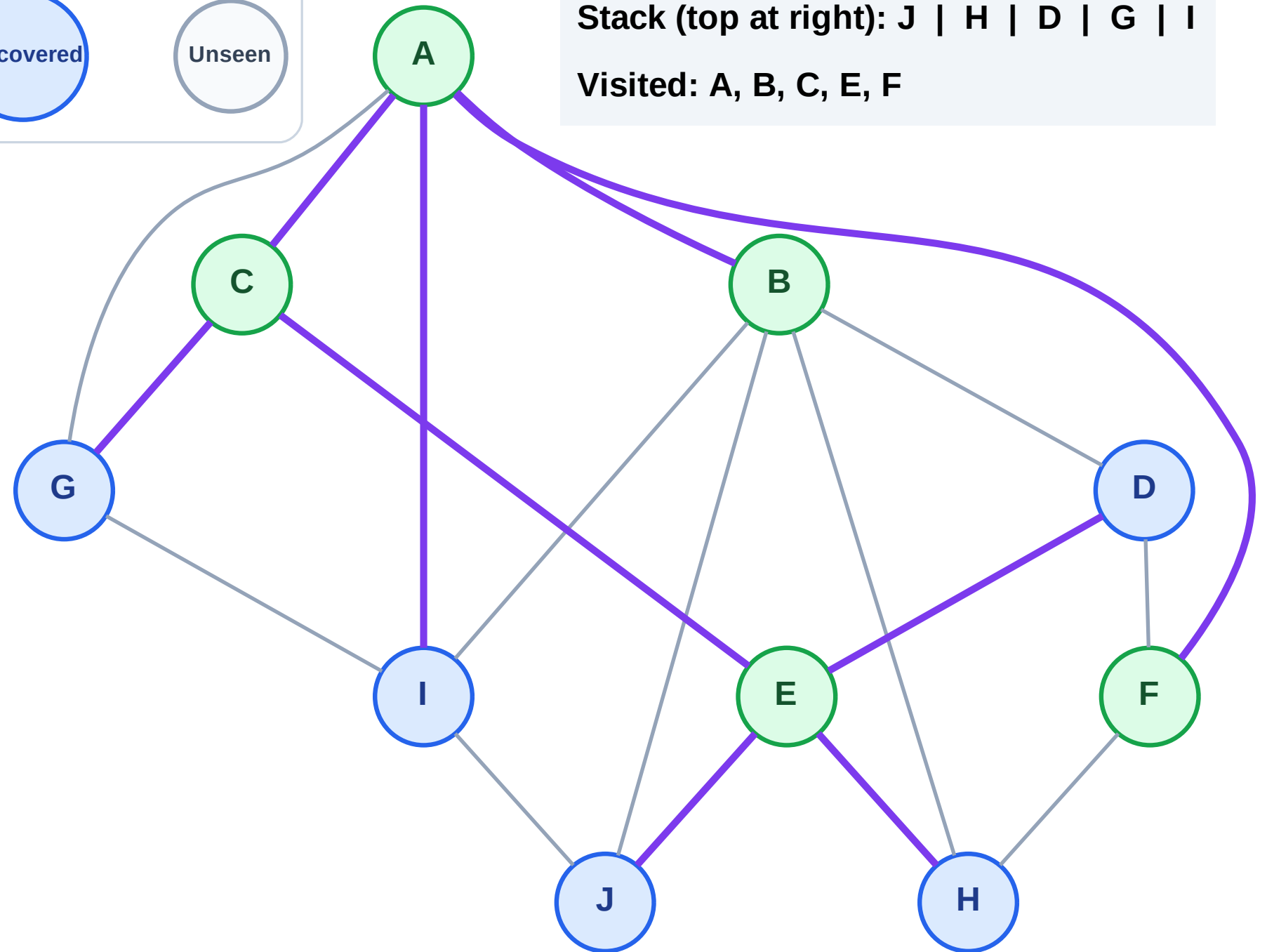
Step 11: No new neighbors from F

Legend



Stack (top at right): J | H | D | G | I

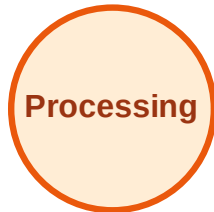
Visited: A, B, C, E, F



DFS Traversal

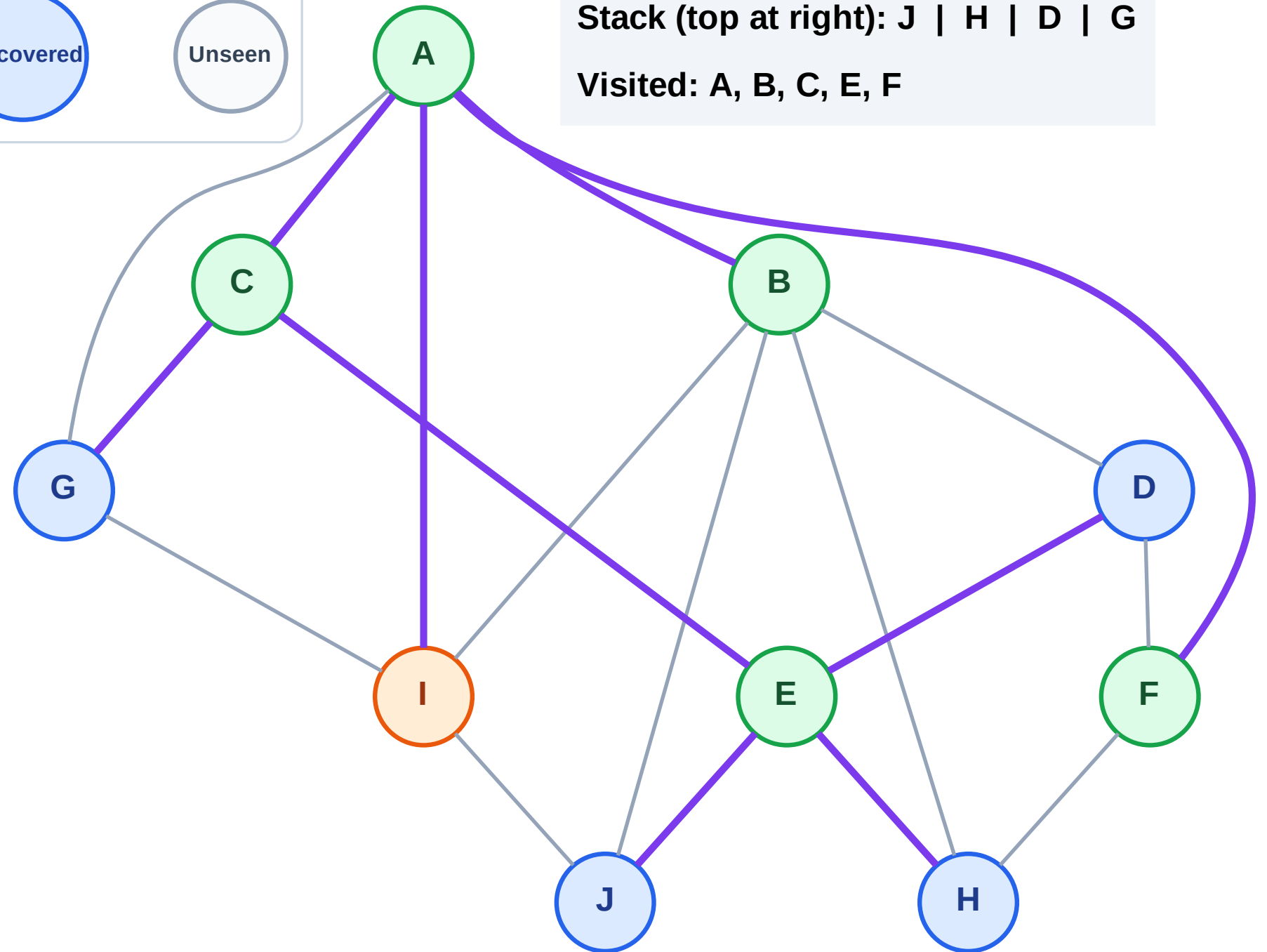
Step 12: Process node I

Legend



Stack (top at right): J | H | D | G

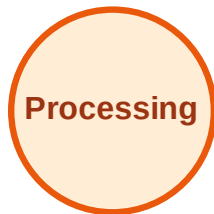
Visited: A, B, C, E, F



DFS Traversal

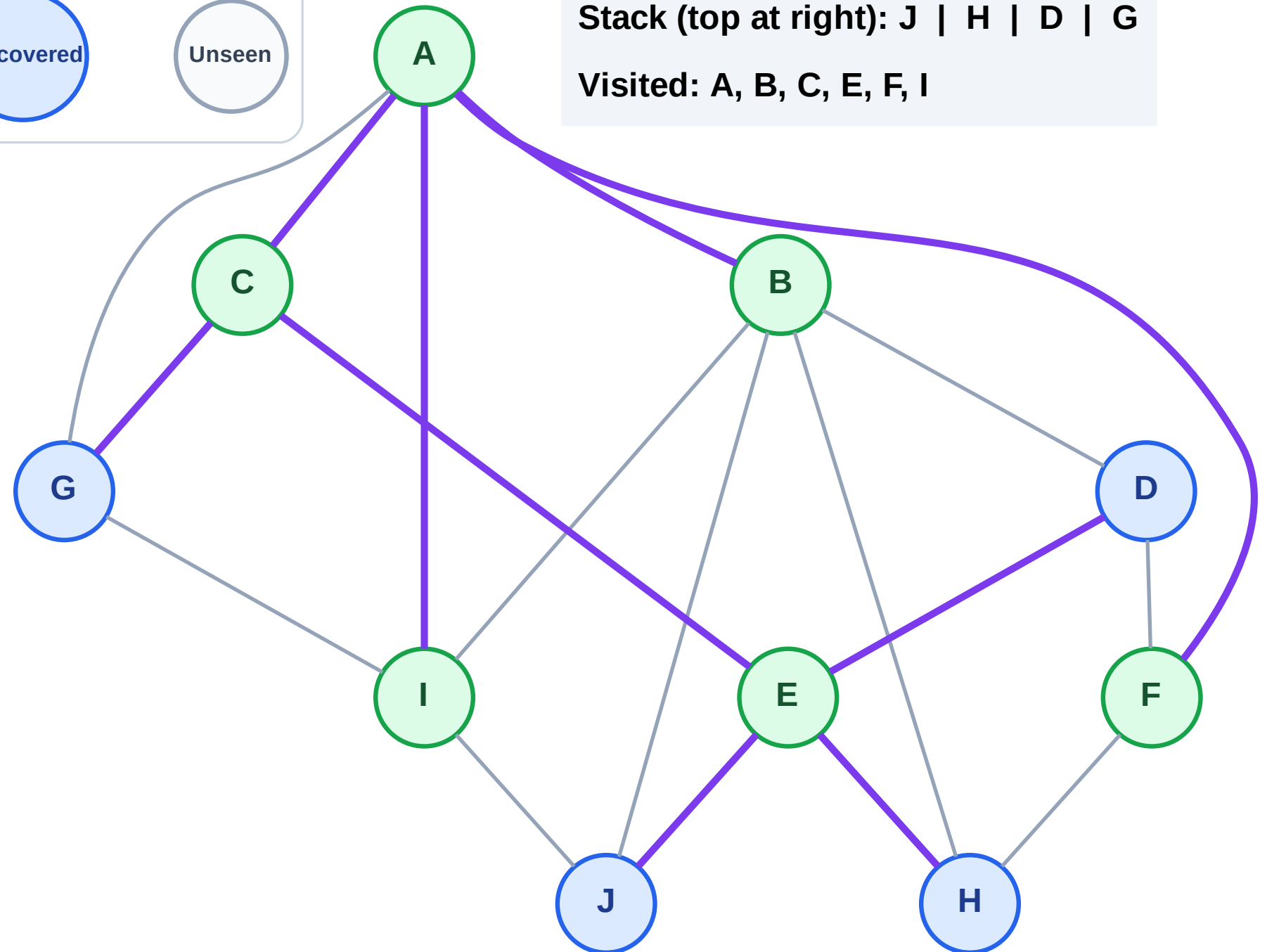
Step 13: No new neighbors from I

Legend



Stack (top at right): J | H | D | G

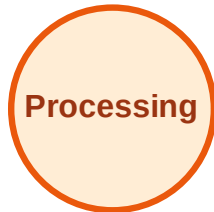
Visited: A, B, C, E, F, I



DFS Traversal

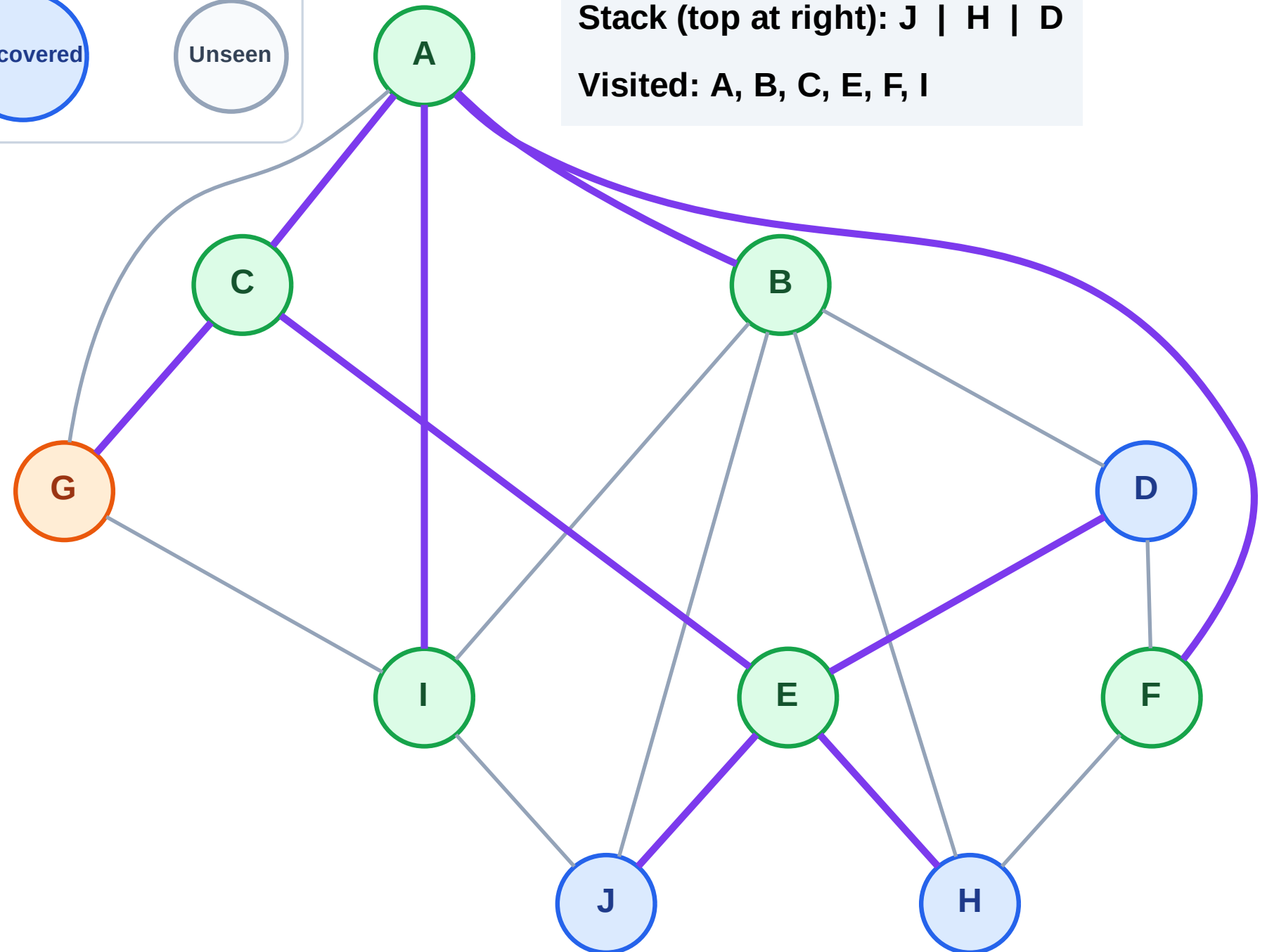
Step 14: Process node G

Legend



Stack (top at right): J | H | D

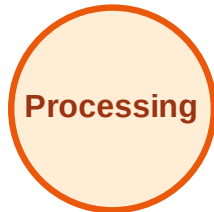
Visited: A, B, C, E, F, I



DFS Traversal

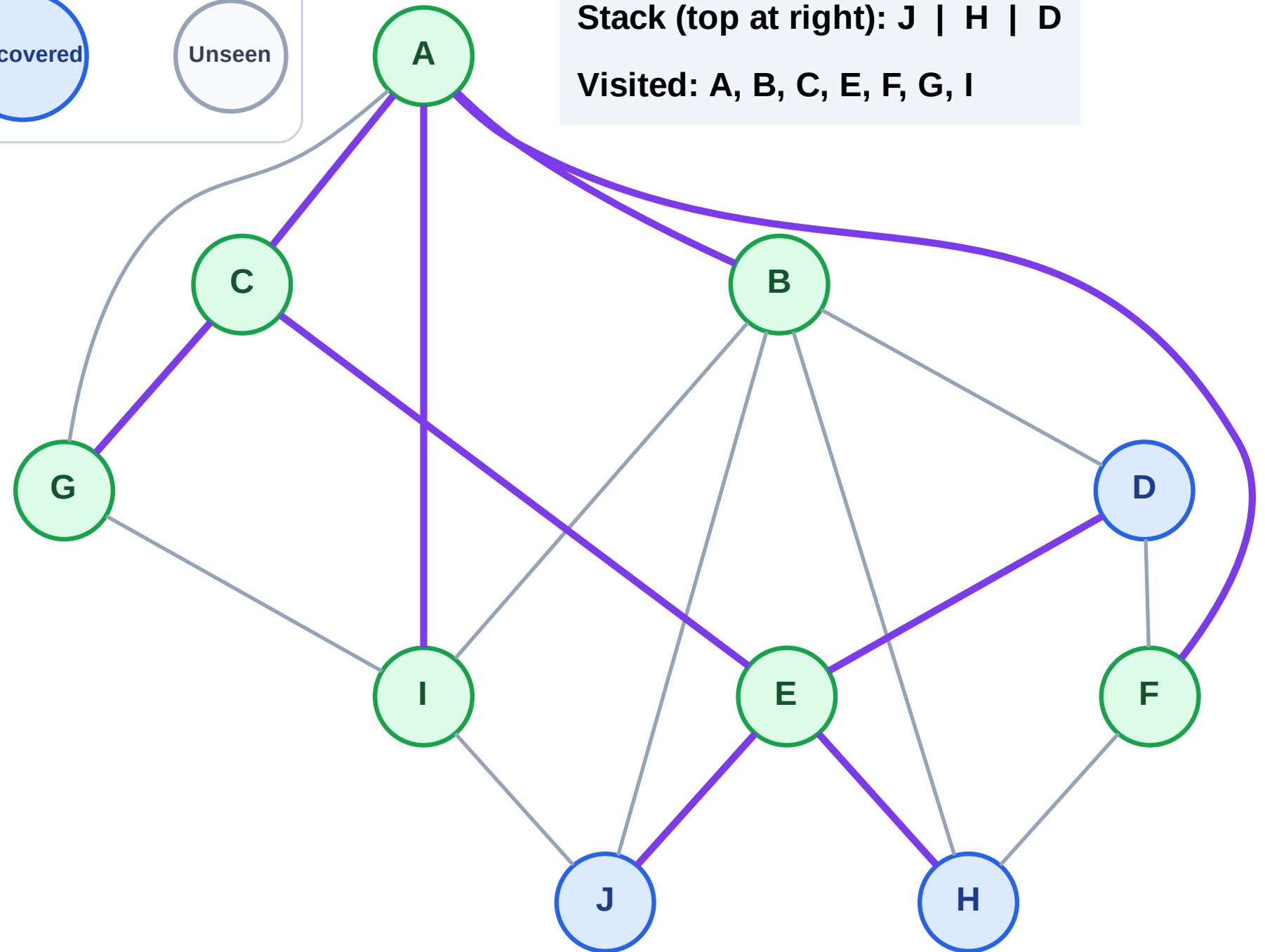
Step 15: No new neighbors from G

Legend



Stack (top at right): J | H | D

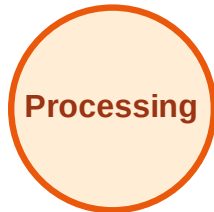
Visited: A, B, C, E, F, G, I



DFS Traversal

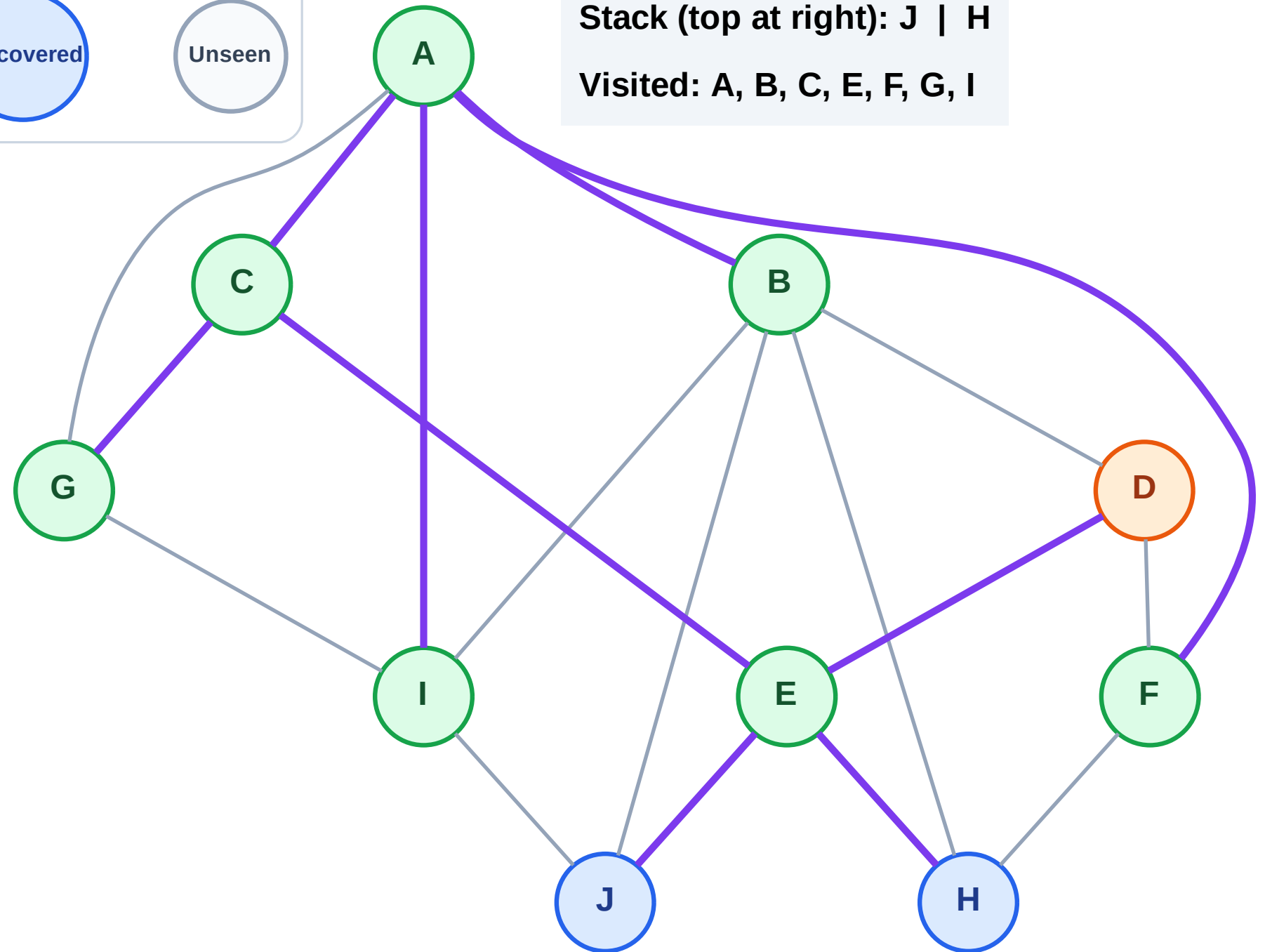
Step 16: Process node D

Legend



Stack (top at right): J | H

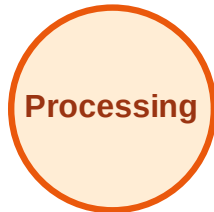
Visited: A, B, C, E, F, G, I



DFS Traversal

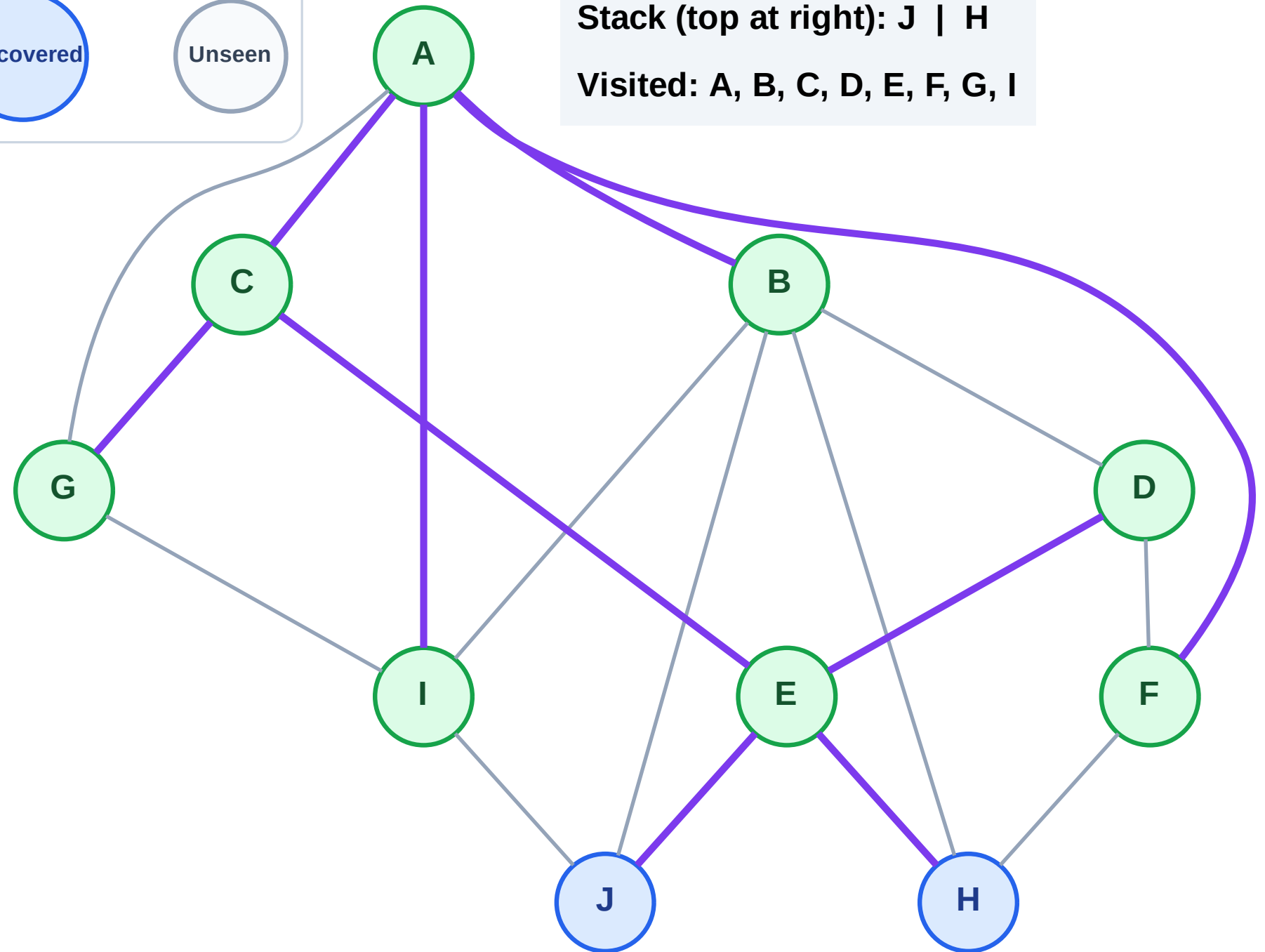
Step 17: No new neighbors from D

Legend



Stack (top at right): J | H

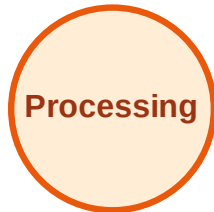
Visited: A, B, C, D, E, F, G, I



DFS Traversal

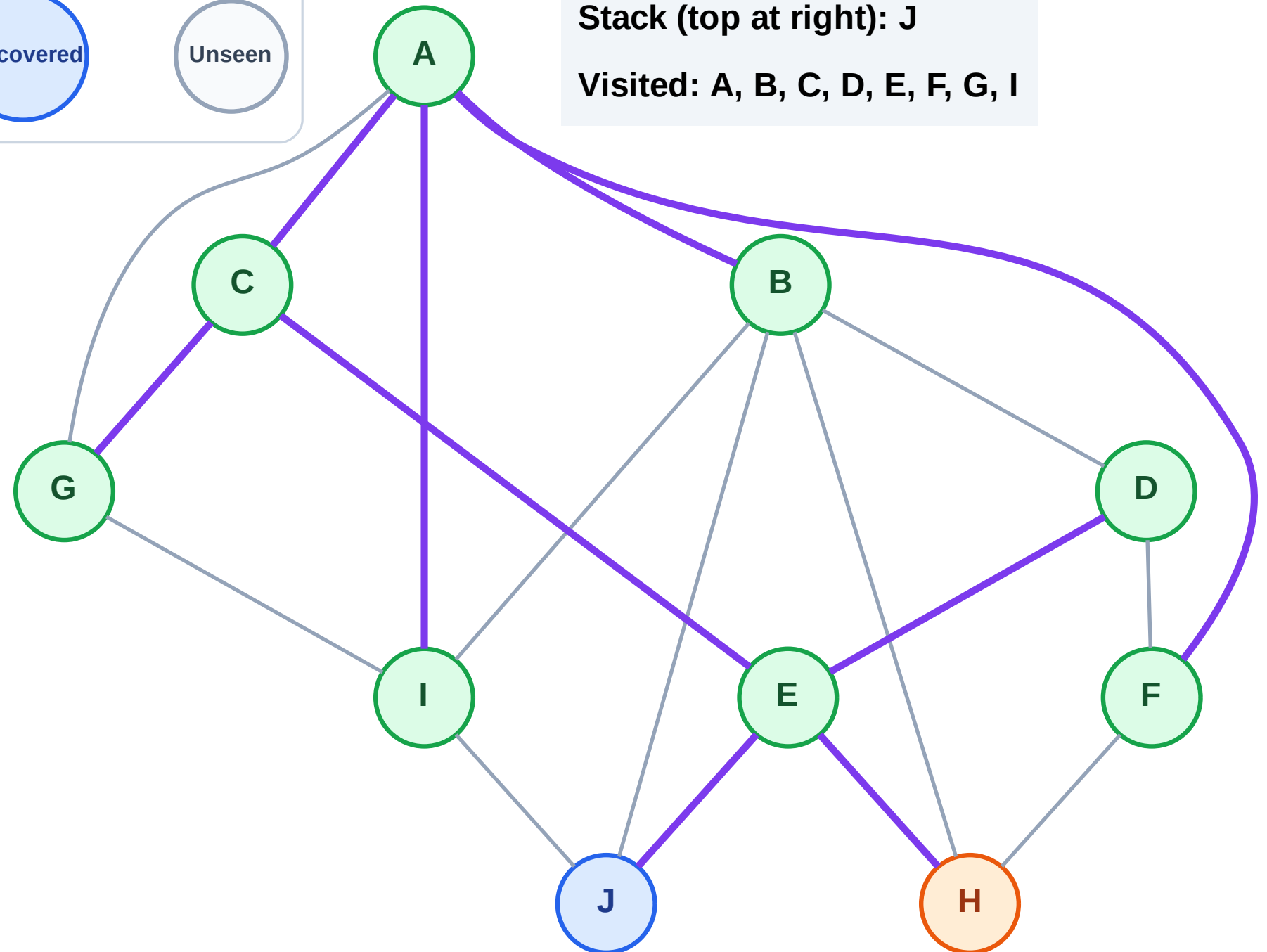
Step 18: Process node H

Legend



Stack (top at right): J

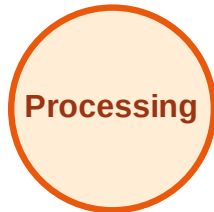
Visited: A, B, C, D, E, F, G, I



DFS Traversal

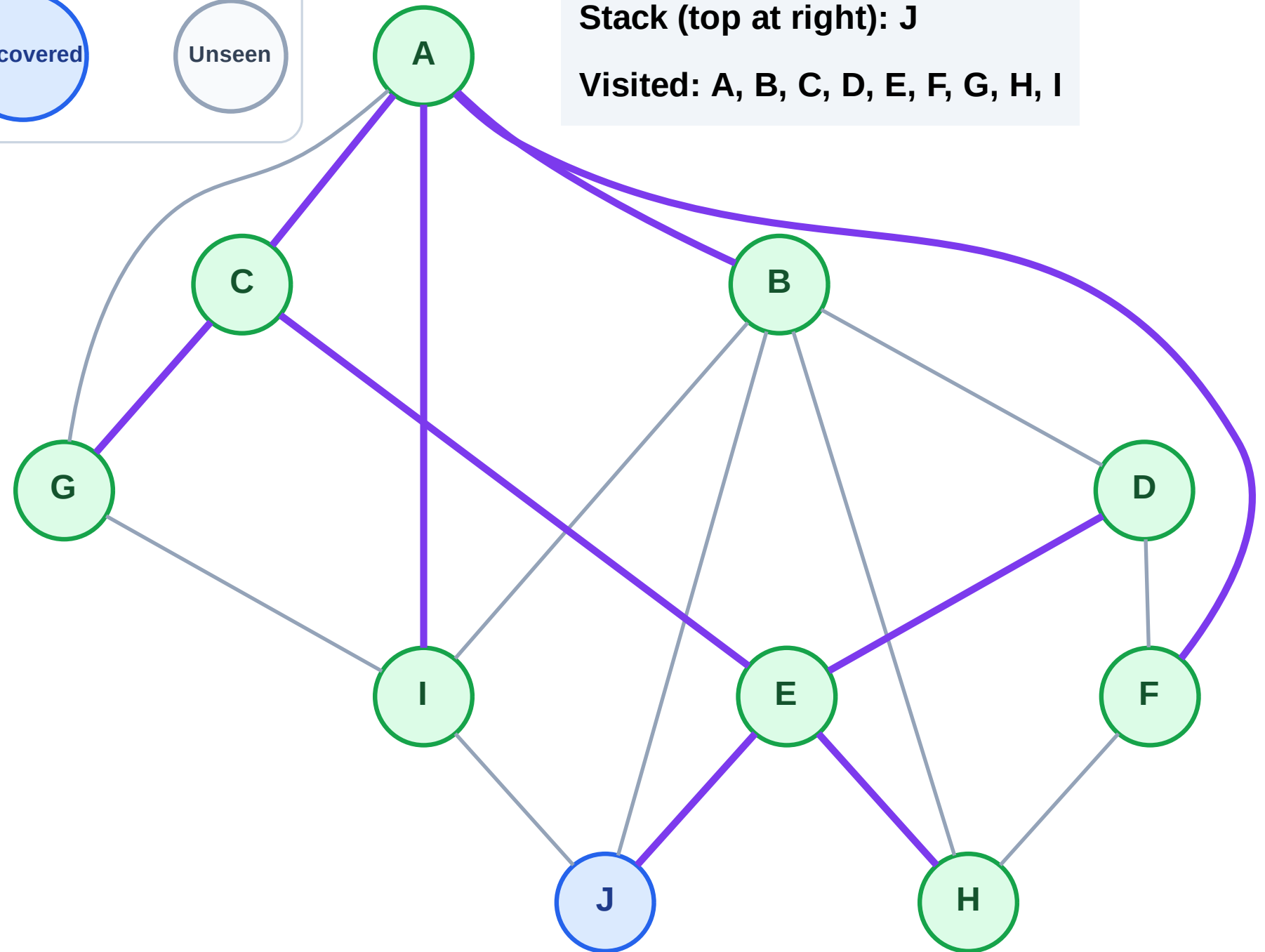
Step 19: No new neighbors from H

Legend



Stack (top at right): J

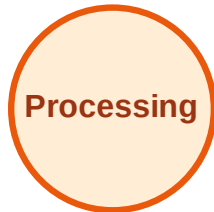
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

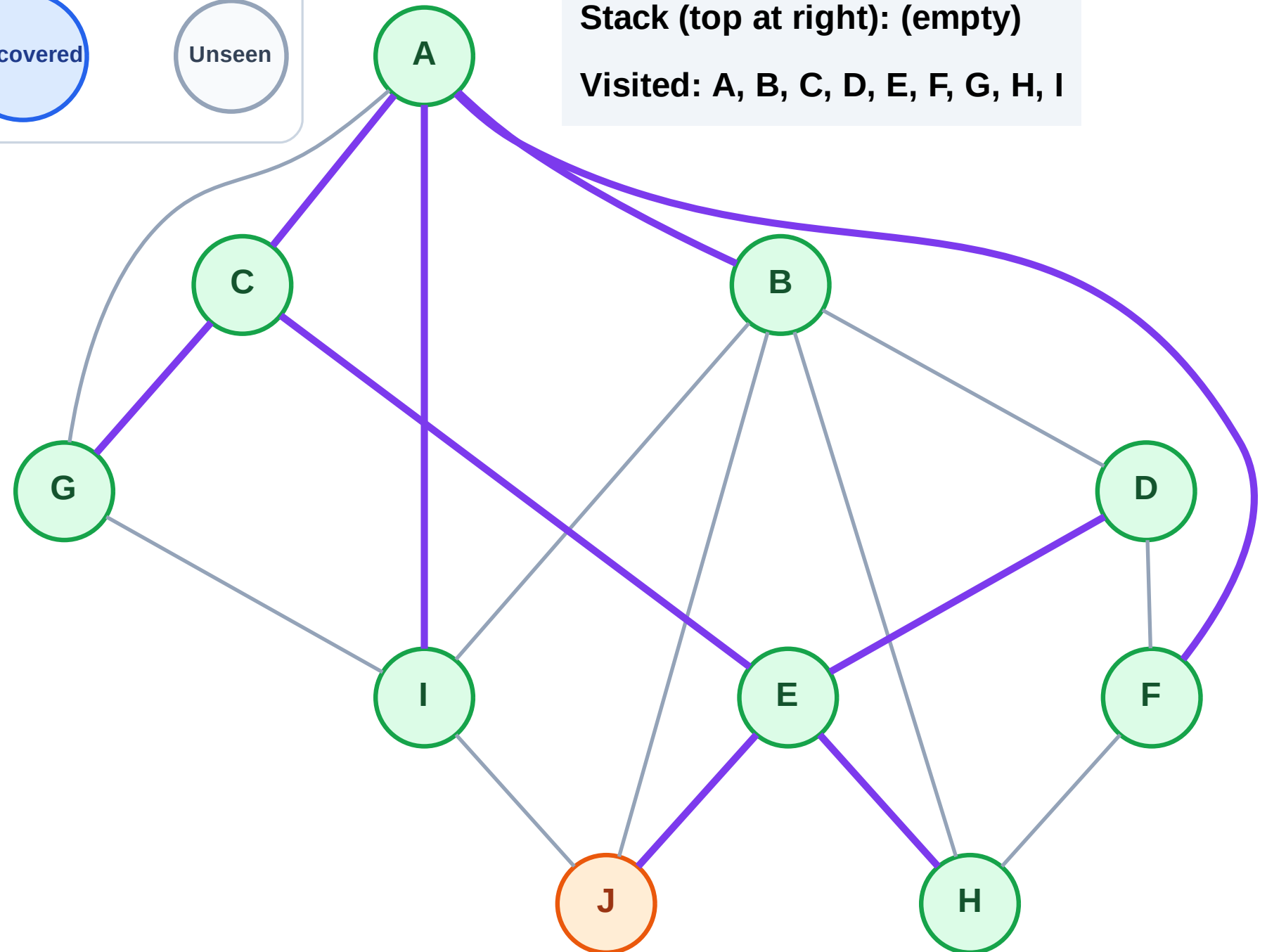
Step 20: Process node J

Legend



Stack (top at right): (empty)

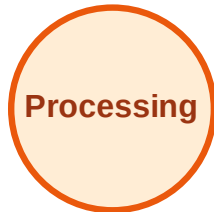
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

